THE LONG-TERM CONSEQUENCES OF FREE SCHOOL CHOICE

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Abstract
I study the long-term consequences of an effective free school choice program that targeted disadvantaged students in Israel two decades ago. I show that the program led to significant gains in post-secondary education through increased enrolment in academic and teachers' colleges without any increase in enrolment in research universities. Free school choice also increased earnings at the adulthood of treated students. Male students had much larger improvements in college schooling and labor market outcomes. Female students, however, experienced higher increases in marriage and fertility rates, which most likely interfered with their schooling and labor market outcomes. (JEL: J24)

1. Introduction

The evaluation of educational programs and interventions has focused on short-term outcomes, primarily standardized test scores, as a success measure. However, understanding that education aims to enhance lifetime well-being, attention has moved recently to long-term consequences at adulthood. In light of the increasing economic returns to higher education, the initial focus has been on post-secondary

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attainment (Acemoglu and Autor 2010; Heckman and LaFontaine 2010). Garces et al. (2002), Ludwig and Miller (2007), and Deming (2009) studied the long-term benefits of Head Start; Schweinhart et al. (2005) examined the long-term effect of the Perry Preschool program; Chetty et al. (2011) studied the impact of kindergarten classroom on earnings in early adulthood; Dustmann et al. (2012) examined the effect of high school quality on completed schooling and labor market outcomes; Dynarski et al. (2013) examined the effect of smaller classes in primary school on college entry, college choice, and degree completion; Deming et al. (2013) studied the impact of accountability pressure in high schools on post-secondary attainment and earnings; Chetty et al. (2014a,b) examined the earnings consequences of primary and middle school teacher quality; and Deming, Billings, and Rockoff (2014) studied the impact of the end of race-based busing on college attainment and young adult crime. Ebenstien, Lavy and Roth (2016) examined the effect of air quality and polution during high school on post-secondary schooling and labor market outcomes; Lavy (2020b) studied the effect of teachers' pay for performance program on students' adulthood outcomes.

These studies' common goal is to determine which interventions are more effective in improving long-term outcomes, but the scope of educational interventions studied is still limited, and much remains to be unraveled. This paper explores the long-term consequences of free school choice offered to primary school students at the transition to secondary schools. The main question I address here is whether the effects of free school choice persist beyond attainment and test scores in high school and lead to long-term enhancements to human capital and well-being. I do so by studying a school choice experiment conducted two and a half decades ago in the city of Tel-Aviv, Israel. The program had positive short- and medium-term effects on cognitive outcomes and schooling attainment during middle and high school and improved students' social skills (Lavy 2010). In this paper, I study whether the free school choice among public schools had long-term effects on social and economic outcomes. This paper provides the first evidence of links between school choice, students' employment and earnings, and social outcomes in adulthood. I examine the impact on various types of post-secondary schooling that vary by quality, employment, and earnings at 11–13 years after high school graduation, almost 20 years after students could exercise free school choice. I also examine several potential mechanisms for these effects.

I observe students' outcomes every year from high school graduation (2000–2001) until age 30–32 (in 2014). Thus, I can estimate the treatment effects for every year and trace the dynamic evolution of the program effect. The evidence shows that school choice increased post-secondary schooling. Treated students are 5.7 percentage points more likely to enroll in post-secondary education and complete an additional fifth year of college than students in the control group. These effect sizes reflect a 13% increase relative to pre-program means and are similar in magnitude to the program's effect on the end of high school matriculation outcomes. The increase in post-secondary schooling mainly reflects an increase in academic (as opposed to vocational) education through increased enrolment in academic and teachers' colleges, but any increase in enrolment in research universities. This is not a surprising result since those affected by the program are marginal students from low socioeconomic backgrounds who would probably not enroll in any academic post-secondary schooling if not for the school choice program.

It is important to note that these results are in general equilibrium because those affected by the experiment are a very small proportion of their cohort. Therefore, the expansion in post-secondary schooling in the treated sample is not at the expense of others who could have been "crowded out" by the new demand for higher education. Furthermore, these effects occurred during a period of expansion of the supply of academic colleges in Israel. Otherwise, the general equilibrium effect's concern will have to be addressed in the context of at-scale implementation of such a school choice program.

Alongside these gains in post-secondary schooling, average annual earnings among treated students 11–13 years after high school graduation increased by 7%–8% relative to the control group mean. These gains are due to improvements in high school outcomes (composite matriculation score, matriculation diploma, number of matriculation subjects at the honors level) and post-secondary schooling attainment, highly correlated with labor market earnings.

Finally, I find that the school choice program did not affect the marriage rate and age of marriage, but it delayed half a year the age of having the first child. However, this finding disguises considerable heterogeneity in effect on marital outcomes by gender, with significant effects on females and no effect on males.

Exploring potential mechanisms for these effects, I find no evidence of sorting of high-achieving students to high-achieving out-of-district schools; namely, the choice program did not cause a "cream-skimming" effect in the form of higher enrolment rate (take-up rate) of strong students relative to weak students in high-achieving schools. However, district 9 students through school choice were able to improve on average the quality of their peers and the quality of the secondary school they attended. This was achieved mainly by better student–school matching and also by a gain in school quality. The larger was the latter, the bigger was the short- and long-term effects of the choice program.

The lessons learned from this analysis are relevant to other educational settings in developed countries. Both the high school system in Israel and its high-stakes exit exams are similar to those in other countries. Importantly, variants of the school choice program studied here have been implemented in developed countries.[1] The Tel-Aviv reform structure also resembles many current programs in US schooling districts. Like in Tel-Aviv, free choice has replaced zoning and cross-district busing.

1. For example, school choice in Sweden and, in particular, in Metropolitan Stockholm provides since the mid-1990s relatively disadvantaged children with better educational opportunities, as students can choose a school within or outside their respective catchment area (Wondratschek, Edmark, and Frolich 2013; Böhlmark, Holmlund, and Lindahl 2015). In the United Kingdom, a national program of parental choice of schools was established, particularly by the Education Reform Act 1988, and through subsequent laws and amending legislation. Consequently, all public and state schools are "choice schools", and all families have the right to choose any school, within or outside their local education authority, and schools cannot anymore refuse entry to any student until reaching enrolment capacity (Fitz and Taylor 2003). Both these programs have many features that are similar to those of the Tel-Aviv program.

In the United States, free choice was introduced under a court order that terminated race-based busing. In contrast, in Tel-Aviv, the choice reform was an initiative of the city school authorities and parents' organizations.[2]

Another essential advantage of the evidence presented in this paper is that school choice can be directly implemented as a public policy. At the same time, most recent studies of longer-term outcomes cover measures that are not as easily influenced by policy, such as school or teacher quality.

There is little causal evidence on school choice's long-term effect, even though it is a controversial policy. The earlier studies examined the short-term effects of voucher programs (e.g. Rouse 1998; Cullen, Jacob, and Levitt 2006). More recently, Chingos and Peterson (2013) report an experiment that offered a private-school voucher to low-income families. Overall, this study reports no significant effects on college enrolment of the voucher offer. Still, they estimate large significant impacts for African-American students and smaller but not statistically significant for Hispanic students. In recent years, the focus has shifted to short- and long-term effects of open enrolment-free school choice programs, particularly on the effects of open enrolment-free school choice programs, particularly on misbehavior and crime post-secondary schooling attainment. Demming (2011) examines the public school choice lotteries in the Charlotte-Mecklenburg school district and finds that seven years after random assignment, lottery winners had significantly reduced crime rates (less severe crimes and fewer days incarcerated). Demming et al. (2013) study the same program in Charlotte-Mecklenburg and find a significant overall increase in college attainment among lottery winners who attend their first-choice school. Wondratschek, Edmark, and Frolich (2013) study the short- and long-term effects of Sweden's 1992 school choice reform and find that it had very small positive effects on marks at the end of compulsory schooling. Still, it had no effects on university education, employment, criminal activity, and health at age 25.

The remainder of this paper is as follows. In Section 2, I describe the Tel-Aviv School Choice Program (TASCP). In Section 3, I present the data, and in Section 4, I outline the identification and econometric model. In Section 5, I present the results, and in Section 6, some conclusions.

2. Background

The analysis of the short- and medium-term effects, presented in Lavy (2010), indicated that the TASCP reduced the dropout rate from grade 7 to 12 by 7.3–8.4 percentage points (a 32% decline), increased the matriculation rate by 8 percentage points (a 25% increase), and increased the average score in the *Bagrut* exams by almost 7 points (a 12% increase). It also improved the quality of schooling as the number

2. An example is the North Carolina Charlotte-Mecklenburg School Board vote to terminate school busing and move forward with district-wide open enrolment for the 2002–2003 school year (for more details, see Deming et al. 2014).

of *Bagrut* credit units increased by 2 (an 11% increase), and the number of credits in science subjects increased by one-third of a unit (a 13% increase). The number of *Bagrut* subjects studied at the honors level increased by 0.3 (a 13% increase). In the following, I summarize the TASCP's main features and then examine whether these gains were translated into economically meaningful adulthood improvements.

2.1. The Tel-Aviv School Choice Program

In May 1994, the Israeli Ministry of Education approved a two-year trial of the TASCP in district 9 (see Map 1 in the Online Appendix). It was the first school choice program in the country since the 1968 education reform, which enacted compulsory integration in grades 7–9.[3] TASCP responded to parents' dissatisfaction with students' outcomes and the severe lack of school choice. Its objectives were to give disadvantaged students access to better schools, facilitate a better match between students and schools, and motivate school productivity improvements through competition. Schooling district 9 included 16 public primary schools—12 secular and 4 religious. Until 1994 the graduates of five of these secular primary schools were bused to one of the five secondary schools in districts 1–5 in north Tel-Aviv (about 36% of the districts' pupils), and a few more of the districts' pupils (5%) were enrolled in charter schools outside the district. The seven other secular primary school graduates were assigned to one of the three secondary schools within district 9.[4]

In May 1994, the Tel-Aviv Education Board announced that as of September 1994, this system would be replaced by free choice for the incoming seventh graders, while older cohorts in the district would continue with the old system. The program's main features were as follows. (1) It allowed students to choose their middle–high school from a given set of five schools that included schools from three outside districts (in districts 1–3, and these were the same schools to which students were bused before the program) and two in-district schools. The choice set varied by primary school, and students were asked to rank their preferences among the five schools in their choice set. The city opened information centers and ran workshops for parents and pupils, and high schools held open days to provide information about the choice program for the incoming seventh-grade cohort. (2) In the event of excess demand for a particular school, students were assigned to schools by a city school authority committee to maintain a socioeconomic balance. Siblings in the same school and school capacity were also used as criteria to balance enrolment. (3) Students were guaranteed that at

3. The 1968 reform established a three-tier structure of schooling: primary (grades 1–6), middle (7–9), and high school (10–12). The reform established neighborhood school zoning as the basis of primary enrolment and of the integration and busing of students out of their neighborhoods in middle school. In Tel-Aviv, most middle schools were part of six-year high schools, and there were several high schools that only offered the higher grades (10–12).

4. These schools were located on the same campus, but they were very different in terms of their curriculum and programs offered to students. For example, one included low- and high-tech vocational schooling.

least one friend from their list of nominated friends would attend the same school. (4) Schools were forbidden from any form of discrimination in accepting students—neither geographic nor merit-based. (5) A decision accompanied the choice program that all the city's post-primary schools would be six-grade structures that included the middle (7–9) and higher grades (10–12) as part of the same school. Most of the city's post-primary schools were already structured this way, and only four schools had to be expanded to include middle-school grades. (6) Each school was given additional autonomy, and task forces were set up to help schools develop their identity and improve their administrative and pedagogical processes. (7) Schools that enjoyed expanded enrolment gained more resources as the school budget was based on enrolment.

Before the implementation of the program, the following steps were taken.School administrators were engaged and briefed by the municipality's education administration. Parents and students were given information on the program and the choices available to them in a series of open meetings, written material distributed, and "open days" that each school held for prospective students. An information center was set up to answer questions and provide additional information through meetings and written material. A significant step in the preparation process was opening a new school in district 9, the second school. It opened in the 1993 school year, and so we can observe it for only one academic year before the program. A third school was also opened the following year, but it had low enrolment from the start and was closed a few years later.

According to city council protocols dating from 1995[5] and also based on other secondary sources, over 90% of students received their first choice, the rest getting their second choice.[6] Interestingly, both the protocols and the secondary sources reveal that in the aggregate, this control system was used to increase the number of students studying in Northern schools—by 15%, relative to the number that would have studied there had there been no control of student choice. Levy, Levy, and Libman (1998) provide details on the number of students who were denied their first choice in in-district schools, based on administrative records that I do not have access to. These numbers demonstrate that almost all students who did not get their first choice chose the new school in district 9. The municipality employed its control and regulation systems only for those students who ranked that school as their first choice. Some of these adjustments were later overturned in an appeal process, though many of the appeals were for students who had received their first choice—possibly indicating that some students used this process to "change their minds".

An essential element in the new program was the expansion of the supply of middle school classes: four high schools, two in district 9 and two in the city's northern districts, which had previously only offered the higher grades (10–12), were expanded to provide middle school grades at the commencement of the reform. In a later section of this paper, I discuss why this change in the structure of secondary

5. Tel-Aviv council protocol dated March 19, 1995. These figures are met with some skepticism from a vocal opponent of the program in the city council.

6. The Tel-Aviv Educational Authority (1999).

schools in Tel-Aviv does not confound my estimates of the school choice program's effect. Beyond the fact that over two-thirds of the secondary schools in the city had already a six-grade structure, I also highlight and discuss the fact that two of the three control groups that I use for identification implemented a similar change in their secondary schools' structure around the same time that it was done in Tel-Aviv.

Over time, the choice program led to the expansion of some high schools and others' contraction. (One school was even closed due to declining enrolment.) Enrolment in the city's schools was also affected by the stricter enforcement of the Ministry's rule that pupils could not attend schools outside of Tel-Aviv. Because school budgets were determined according to enrolment, schools that expanded enrolment gained more resources.

The change to a six-grade structure that includes the middle (7–9) and higher grades (10–12) as part of the same school allowed the city to cancel the admission process at the end of the ninth grade and to introduce the concept of "persistence" whereby students were automatically enrolled in the tenth grade in the school in which they had completed their middle school education. This important component of the school system's reorganization in Tel-Aviv strongly limited schools' ability to select students for their higher grades based on academic performance. The explicit default became that pupils could remain in the school they chose in the seventh grade for the duration of their secondary education. To overcome this default option, a school had to gain explicit approval from a particular city committee: consent was only given in cases where pupils displayed severe behavioral problems and never on the grounds of poor academic performance. This policy change most likely explains a large part of the dramatic decline in the pupil transfer rate in the ninth grade, from about 50% before the choice program to about 15% following it.

The same school choice program was rolled in gradually in the other school districts of Tel-Aviv. First, it was expanded in 1996 to district 8, then in 1998 to district 7, and in 1999 to the rest of the city (Tel-Aviv Educational Authority 2001).

The school choice program in Tel-Aviv was implemented relatively earlier than most choice programs in other countries, but it shared many similar elements. Notably, like the Charlotte-Mecklenburg school choice program of 2002 and other similar programs in the United States, it replaced a school busing regime that served socioeconomic integration in schools. For the first time in Israel, children had a free choice among public schools and could opt out of neighborhood schools.[7]

7. The Charlotte-Mecklenburg, North Carolina, school district (CMS) includes Mecklenburg County, which includes both the inner city areas of Charlotte and its suburbs. Following a 1971 Supreme Court ruling (*Swann v. Charlotte-Mecklenburg Board of Education*) and for over 30 years, CMS schools bused students across the district to achieve racial desegregation. In 2001, this busing plan was terminated when the CMS School Board voted to move forward with districtwide open enrollment for the 2002–2003 school year (see details in Deming et al. 2014). Similar termination of school busing and the introduction of free choice among public schools were adopted over the years in many other school districts in the United States. The CMS and the Tel-Aviv choice programs share other similar features, for example, the submission of a list of school choices for each child, listed in order of preference, extensive information campaign to encourage parents to submit choice forms, preparation of comprehensive booklets with information about each school, informing parents that school choice forms were required to receive a school assignment in the subsequent year, and guaranteeing to most students first choice. However, when demand exceeded

2.2. *The Israeli High School Matriculation System*

When entering high school (tenth grade), students choose to enroll in the academic or non-academic track. Students enrolled in the academic track receive a matriculation certificate (*Bagrut*) if they pass a series of national exams in core and elective subjects between the tenth and twelfth grades. Students choose to be tested at various proficiency levels, with each test awarding one to five credit units per subject, depending on difficulty. Advanced level subjects are taken at a level of four or five credit units; a minimum of 20 credit units is required to qualify for a *Bagrut* certificate. About 52% of all high school seniors received a *Bagrut* in the 1999 and 2000 cohorts. The *Bagrut* is a prerequisite for university admission, and receiving it is an economically crucial educational milestone. For more details on the Israeli high school system, see Abramitzky and Lavy (2014).

3. The Data

In this study, I use data from administrative files for students in primary schools who were enrolled into the school choice program, pre-treatment (sixth graders in 1992 and 1993) and post-treatment (sixth graders in 1994) cohorts, and, similarly, repeat it for the control schools as well. The students in the sample completed high school between 1999 and 2001, and in 2013 they are adults, aged 29–31. I use several panel data sets available from Israel's National Insurance Institute (NII). The NII is responsible for social security and mandatory health insurance in Israel. NII allows restricted access to these data in their protected research lab.

The underlying data sources include (1) the population registry data (a national ID number that also appears in all the data sets described in the following and enables matching and merging the files at the individual level), which contain information on marital status, the number of children, and their birth dates; (2) NII records of post-secondary enrolment from 2000 through 2013 based on annual reports submitted to NII by all post-secondary education institutions, from which we calculated the number of years of post-secondary schooling[8]; (3) Israel Tax Authority information on the income and earnings of employees and self-employed individuals for the period 2000–2014; and (4) NII records on unemployment benefits, marriage, and fertility for the period 2009–2012.

supply, admission was allocated in CMS by lottery, while in Tel-Aviv such circumstances were resolved by an assignment committee, as explained above. In both programs, siblings of currently enrolled children received guaranteed access, free transportation was provided, and the school authority expanded capacity at schools where it anticipated high demand in an attempt to give everyone their first choice.

8. The NII, which is responsible for the mandatory health insurance tax in Israel, tracks post-secondary enrolment because students pay a lower health insurance tax rate. Post-secondary schools are, therefore, required to send a list of enrolled students to the NII every year. For the purposes of our project, the NII Research and Planning Division constructed an extract containing the 2001–2013 enrolment status of students in our study.

The NII matched and merged these files using the individual level national ID number. The matching is perfect without loss of observations. NII linked this merged file to students' background data in Lavy (2009) to study the teachers' incentive experiment on high school academic outcomes. This information comes from the Ministry of Education's administrative records on Israeli primary schools' universe during the 1997–2002 school years. In addition to an individual identifier and a school and class identifier, it also included the following family-background variables: parental schooling, number of siblings, country of birth, date of immigration if born outside of Israel, ethnicity, and a variety of high school achievement measures. This file also included a treatment indicator, school ID, and a cohort of the study. I had restricted access to these data in the NII research lab at the NII headquarters in Jerusalem.

3.1. The Post-High School Academic Schooling System in Israel

The post-high school academic schooling system in Israel includes 7 universities (one of which confers only graduate and Ph.D. degrees) and over 50 colleges that confer academic undergraduate degrees (some of these also give master's degrees).[9] All universities require a *Bagrut* diploma for enrolment. Most academic colleges also require a *Bagrut*, though some look at specific *Bagrut* diploma components without requiring full certification. It generally is more difficult for a given area of study to be admitted to a university than to a college. The national university enrolment rates for the cohort of graduating seniors in 1995 (through 2003) was 27.6% and the rate for academic colleges was 8.5%.[10]

The post-high school outcome variables of interest here are indicators of ever having enrolled in a university or an academic college as of the 2013 school year and the number of years of schooling completed in these two types of academic institutions by this date. We measure these two outcomes for our 1999–2001 twelfth-grade students. Even after accounting for compulsory military service,[11] we expect that most students who enrolled in academic post-high school education, including those who continued beyond the undergraduate level, graduated by the 2013 academic year. Figures 1–3 present evidence about post-secondary education evolution and show very small changes in ever-enrolled rates and years of schooling beyond 30. The similarity in magnitude of the treatment rates and the control group during the most recent years suggests no understatement or "catch-up" elements in measuring the effects of the program on post-secondary education. Data published by the Israeli Central Bureau

9. A 1991 reform sharply increased the supply of post-secondary schooling in Israel by creating publicly funded regional and professional colleges.

10. These data are from the Israel Central Bureau of Statistics, Report on Post-Secondary Schooling of High School Graduates in 1989–1995 (available at: http://www.cbs.gov.il/publications/h_education02/h_education_h.htm).

11. Boys serve for three years and girls for two (longer if they take a commission). Ultra-orthodox Jews are exempt from military service as long as they are enrolled in seminary (Yeshiva); orthodox Jewish girls are exempt upon request; Arabs are exempt, though some volunteer.

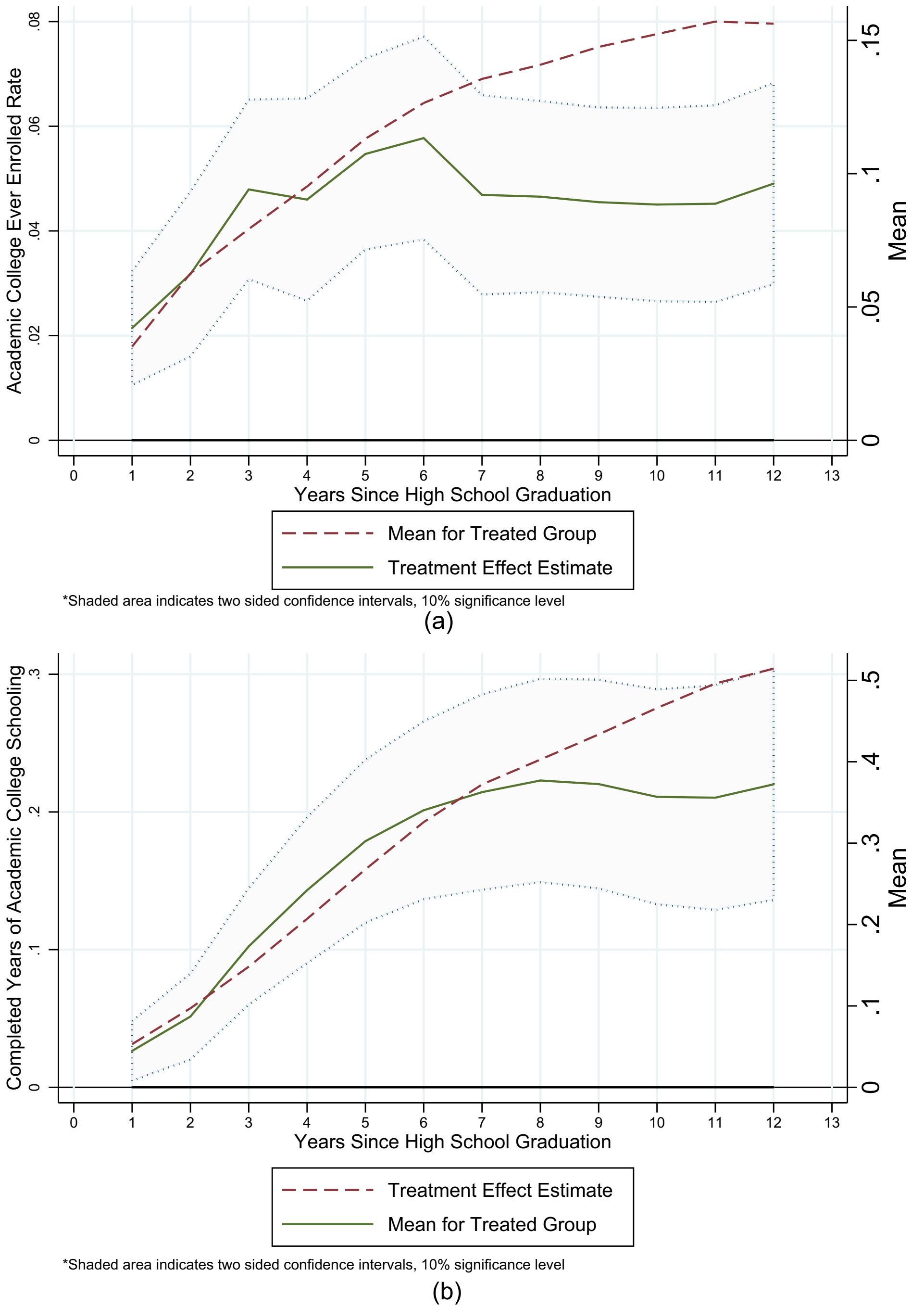
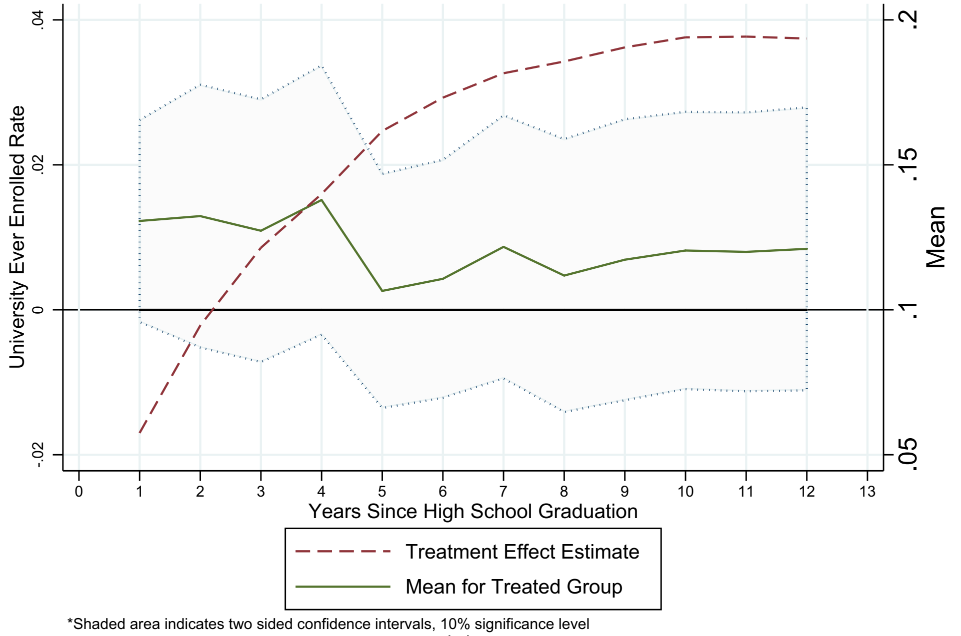
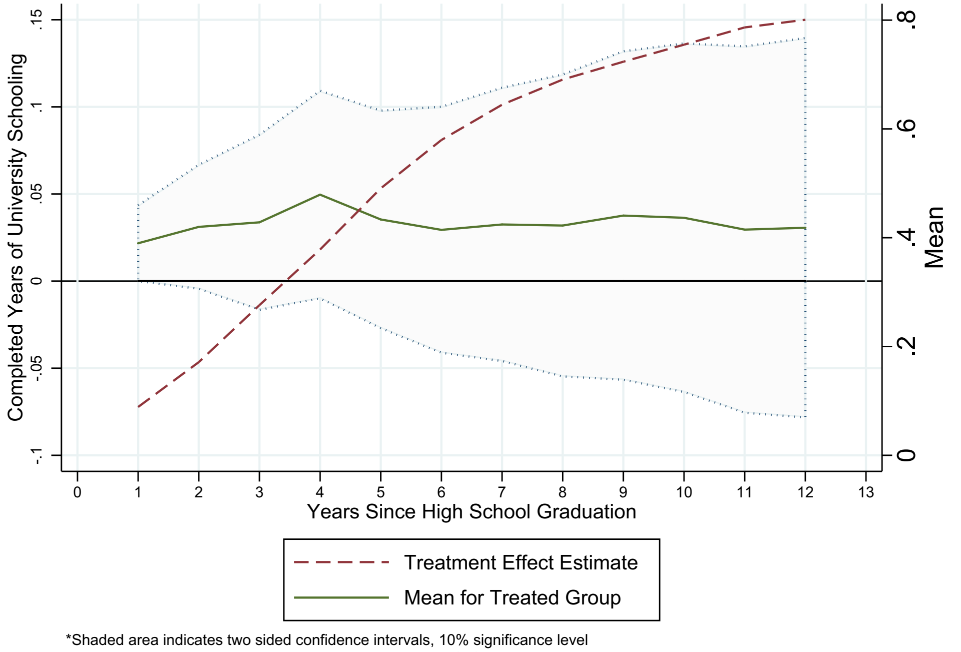


FIGURE 1. (a) Mean and treatment effect: acadmic college enrolment; (b) Mean and treatment effect: acadmic college years of schooling.



(a)



(b)

FIGURE 2. (a) Mean and treatment effect: university enrolment; (b) Mean and treatment effect: university years of schooling.

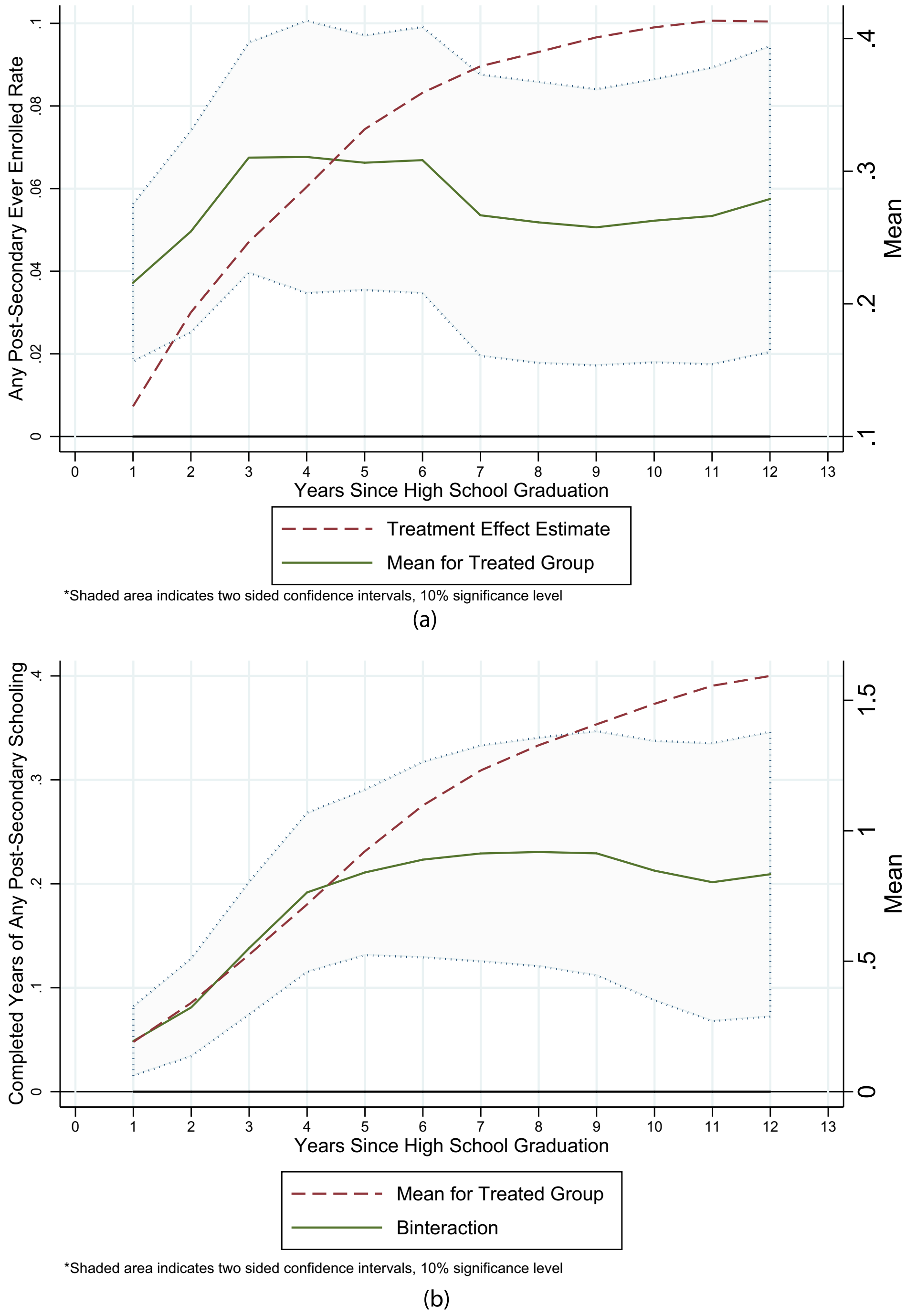


FIGURE 3. (a) Mean and treatment effect: any post-secondaery enrolment; (b) Mean and treatment effect: any post-secondary years of schooling.

of Statistics show similar evidence, for example, in 2016, among bachelor degree students (the appropriate degree level given our population of study), about 13% of all students aged 30 and over (Central Bureau of Statistics 2020). Therefore, the possibility of effects beyond 2013 is less likely.

3.2. Definitions of Outcomes in Adulthood

In this subsection, I describe the outcomes in adulthood for students in the sample. The post-secondary schooling outcomes are also changed for years since graduating high school age differences of the different cohorts included in the sample.

Labor Market Outcome. *Earnings*: Individual earnings data come from the Israel Tax Authority (ITA). Only individuals with non-zero self-employment earnings are required to file tax returns in Israel. Still, the ITA has information on annual gross earnings from salaried and non-salaried employment. They transfer this information, including the number of months of work in a given year, annually, to the NII.

The NII produces an annual series of yearly earnings from salaries and self-employment, and I used this variable for the period 2000–2014. Following NII practice, individuals with a positive (non-zero) number of months of work and zero or missing value for earnings are assigned zero earnings. A total of 14.1% of individuals have zero earnings at age 30–32 in our primary sample of 13,142. I dropped from the sample all observations that are six or more standard deviations (SDs) away from the mean to account for earnings data outliers. Very few observations are dropped from the sample in each of the years, and the results are not qualitatively affected by this sample selection procedure. To account for age differences of the different cohorts in the sample, the employment and earnings outcomes are adjusted for years since graduating high school.

The same earnings data are also available for the students' parents in our sample for the period 2000–2002 and 2008–2012. I compute each parent's average earnings and the household for the period 2000–2002 and use it as an additional control in a robustness check of the evidence presented in this paper. These data were not available for the analysis of the effect of the program on short-term outcomes. *Employment*: An indicator with value 1 for individuals with non-zero number months of work in a given year, 0 otherwise.

Education. Here as well, I measure the outcomes by adjusting for years since graduating high school. *University schooling*: It is an indicator for being enrolled for at least one year in university schooling. Years of university schooling is the number of years of attendance during the period 2000–2013. *Academic college schooling*: It is an indicator of being enrolled for at least one year in any academic college. Years of college schooling is the number of years of attendance.

Personal Status Outcomes. The data on marital status and having children are available only for 2011. Therefore, for these outcomes, we can adjust for years since graduating from high school based on information about marriage dates and children's birth dates.

Marriage. It is an indicator of being married. *Children*: It is an indicator for having at least one child. *Number of children*: It is the number of children.

The NII linked these data to students' background data that I used in Lavy (2010) to study the choice program's effect on high school academic outcomes.[12] This information comes from the Ministry of Education's administrative records on Israeli primary schools' universe during the 1992–1994 school years. In addition to an individual identifier and a school and class identifier, it also included the following family-background variables: parental schooling, number of siblings, country of birth, date of immigration if born outside of Israel, ethnicity, and a variety of high school and *Bagrut* high school achievement measures. This file also included a treatment indicator, school ID, and a cohort of the study. I had restricted access to this data in the NII research lab at the NII headquarters in Jerusalem.

4. Identification and Estimation

In previous work (Lavy 2010), I used the difference-in-differences (DID) and geographical discontinuity (GD) in program placement as two alternative methodologies to estimate the effect of the school choice program on short-term outcomes (dropout rate) and medium-term outcomes (success at the end of high school, six years after the school choice decision, in high-stakes exams). Using a DID methodology, I relied on three alternative comparison groups, which all yielded almost identical evidence regarding the choice program's impact.[13] Therefore, I use this same identification method to estimate the effect of school choice on long-term adulthood outcomes while combining all three comparison groups into one to increase efficiency in the estimation. However, results based on each of these comparison groups separately are in line with the evidence obtained from pooling all three control groups into one large sample. I discuss these results later in the paper and present them in an Online Appendix table.

The GD approach yielded evidence about the choice program's effect on high school outcomes consistent with the DID estimation evidence (Lavy 2010). I use

12. As high school outcomes, I used an indicator of dropping out before completing the twelfth grade, an indicator for matriculation (*Bagrut*) eligibility, credit-weighted average score on the matriculation exams, number of matriculation credits, number of matriculation credits in science subjects, and number of matriculation subjects at the honors level. *Bagrut* eligibility is a prerequisite for admission to higher education in Israel, and the average score on the matriculation exams, number of matriculation credits in science subjects, and number of matriculation subjects at the honors level are used to screen and select students for prestigious universities and sought-after academic programs such as medicine, engineering, and computer science.

13. In Online Appendix Table A.1, I present the mean demographic characteristics of the students in the treatment group and in each of the three alternative control groups used in the DID estimation. This table is a replication of Lavy (2010, Tab. 1).

the GD identification method in this paper as well. I summarize the comparison group used herein for the DID estimation and the comparison group used in the GD estimation. More details about each of them are provided in Lavy (2010).

The comparison group in the DID estimation includes three different control groups. The rationale of the first is the gradual implementation of the program. Therefore, it includes school districts 6–8 in Tel-Aviv, which were enrolled immediately following the two-year experiment in district 9.[14] All three districts are part of South Tel-Aviv, geographically adjacent to or near district 9 (see Map 1 in the Online Appendix). Their population is much more similar to that of district 9 than that of the other school districts in Central and Northern Tel-Aviv (Lavy 2010).

The second comparison group includes two adjacent cities east of Tel-Aviv, which are part of the Dan metropolitan area.[15] District 9 is the city's southeastern neighborhood and is tangential to two neighboring towns: Givataim and Ramat-Gan (referred to as GR, see Map 1 in the Online Appendix). GR have independent and separate education systems and therefore were not part of Tel-Aviv's school choice reform.[16] GR students are very different in mean characteristics from district 9 students (Lavy 2010). However, these differences are very stable as they are similar in 1992 and 1993 as well. Therefore, the pre-program imbalance solution uses pre-and post-program cohorts (panel data) in a DID framework that removes time-invariant heterogeneity across treated and control groups.

Holon is another city in the Dan metropolitan area adjacent to Tel-Aviv (south), and it is very close to district 9. It has three additional attractive features as a potential comparison group: (1) its high school enrolment system before the inception of the TASCP was based on zoning, and it has not changed since; (2) it changed its secondary school structure to a six-grade structure in 1995; and (3) it is more similar to district 9 in its characteristics than the GR group. I use it as a third control group.[17]

Therefore, this paper's primary identification approach is based on a contrast between district 9 and a comparison group that includes these three control groups together, before and after the program was implemented. I use data on pre-and post-program cohorts (panel data) in a DID framework that removes any remaining time-invariant heterogeneity across the treated and control groups since this DID estimation compares two consecutive cohorts. Since the program was implemented

14. Note that since all the schools in districts 1–5 were included in the choice sets of students in district 9, they cannot be included in the comparison group regardless of the date on which they implemented the choice program.

15. The Dan metropolitan area includes ten cities, including Tel-Aviv.

16. The GR high school enrolment system before the inception of the TASCP was based on zoning, and it has not changed since, nor have these cities undergone any other major educational reform since 1994.

17. The fact that three alternative sets of DID estimates, one that is based on a comparison group that has much better characteristics and outcomes (GR) than the treated group, a second that is based on a comparison group that has marginally worse characteristics and outcomes (districts 6–8), and a third that has relatively similar characteristics (Holon), yield exactly the same results is reassuring, given the possibility that the DID estimates are biased because of regression to the mean or due to differential time trends in unobserved heterogeneity between treatment and control.

immediately after it was announced, it is reasonable to assume that the remaining differences were constant within this narrow time range.

However, a concern with this DID approach is that the cohort immediately before the treated cohort that I use as a control group might be affected through spillover effects at the school level. As students of this cohort will be attending the same schools as the treated students, peer effects or competitive effects on school productivity might impact the untreated students. As a robustness check, I re-estimated the econometric models based on using as a control group the cohort that started middle school two years before implementing the program. In this case, it is expected that spillover effects will be smaller. I used a similar approach in Lavy (2010), and the results indicated no spillover effect. I replicated this analysis with the longer-term outcomes (post-secondary schooling and labor market outcomes) and obtained similar evidence of no spillover effects.

As noted previously, I also use a GD in program placement as an alternative identification strategy. Following Black (1999), I limit the sample to observations within a narrow band around the municipal border between district 9 and GR (see Map 2 in the Online Appendix). As shown in Lavy (2010), the physical and other characteristics of the communities within this strip (for example, type and the average size of homes) are identical, as are zoning laws and municipal (type of property) taxes, which are determined by the central government. Presumably, there might still be some differences, such as the mayor's political affiliation. The concern remains then that such remaining differences may confound the effect of the program. As mentioned previously, the use of data on pre-and post-program cohorts in a DID framework will remove such time-invariant heterogeneity across the treatment and control groups. I define this sample based on drawing a symmetric band around the municipal border, 250 or 500 meters on each side. Contrary to the imbalances between district 9 and GR, this GD sample yields better-balanced treatment and control groups. In the analysis of the long-term outcomes, I will use the +/–500 meters' band to have a larger sample for estimation, but it should be noted that the +/–250 meters' band yields similar results.

4.1. Estimation

I first present a controlled comparison of treated and untreated students using samples of pre- and post-treatment cohorts based on the following regression:

$$Y_{ijt} = X_{ijt}\beta + Z_j d + U_{ijt}, \tag{1}$$

where Y_{ijt} is the ith student's outcome in school j and year t; X_{ijt} is a vector of the same student's characteristics; Z_j is the treatment indicator (which equals 1 for district 9 students), and d is the treatment effect. As noted previously, I will first estimate the equation using as a comparison group a sample that includes Tel-Aviv district 6–8 students and GR and Holon students. Then I will also exploit the GD sample (using the +/–500 meters' sample).

Also, I use the before-and-after cross-section data as stacked panel data that permit regression analysis with controls for primary-school fixed effects. Therefore, I will estimate stacked models using three years of cross-section data combined. The treatment indicator Z_{jt} is now defined as the interaction between a dummy for the year 1994 and the district 9 indicator as follows:

$$Y_{ijt} = \mu_j + \pi_t + X_{ijt}\beta + Z_{jt}d + \varepsilon_{ijt}, \tag{2}$$

where μ_j is the primary-school fixed effect and π_t is a year (i.e. 1992, 1993, and 1994) fixed effect. Apart from providing a check on the precision of the 1992–1993 versus 1994 contrast in treatment effects, the introduction of school (fixed) effects also provides an alternative approach to the clustering problem. However, the validity of this control depends on the validity of an additive conditional mean function as a specification for potential outcomes in the absence of treatment.

4.2. Descriptive Statistics

Table 1 presents detailed descriptive statistics of the outcome variables for eleven years since high school graduation for the 1992–1994 cohorts by treatment and control groups, and by treatment and control groups and pre-and post-reform cohorts. Post-secondary enrolment statistics are presented in panel A. The enrolment rate in university schooling in the treatment group for the pre-treatment cohorts (1992 and 1993) is 19.5%, and for the control group, it is 24.5%. The difference is –0.051 (standard error [SE] = 0.012), and is statistically different from zero. The respective enrolment rates in academic colleges are 15.7% and 23.0%; the difference is –0.072 (SE = 0.012) and statistically different from zero.[18,19] The respective means and estimated differences for the post-treatment cohort are presented in columns 4–6. Note that the mean difference in university enrolment did not change much, while the mean difference in the theoretical college enrolment rate declined to –0.033 from –0.072. The difference between these two differences (which is a simple uncontrolled DID estimate), 0.049, will be shown to be very close to the controlled DID estimate that I will present in the next section.

Summary statistics on completed years of schooling are presented in panel B. The average number of years of university completed eleven years after high school graduation in the treatment group's pre-reform cohorts is 0.788. In the control group, it is 1.053. The difference is –0.264 (SE = 0.058). The respective means for years of college education are 0.495 and 0.810, and the difference is –0.315 (SE = 0.045). As expected, this evidence suggests that the treatment–control imbalance at baseline in both types of post-secondary education is in favor of the control group.

A similar treatment–control comparison based on the post-treatment cohort (1994) reveals that the university attendance gap remained unchanged. The gap in academic colleges' attendance almost eliminated, and the remaining difference became statistically negligible. The implied simple DID estimate for college enrolment is 0.185 years, and for university, it is –0.070 years. These dynamic changes suggest

18. Note that very few students ever enrol in more than one type of post-secondary educational institution.

19. The means for the whole cohort (82,500 students) are 24.0% in universities and 24.0% in academic colleges.

TABLE 1. Descriptive statistics and pre- and post-treatment-control comparison of means of post-secondary schooling outcomes, employment and earnings, and personal status outcomes (eleven years since high school graduation).

	Pre: 92 and 93 cohorts			Post: 94 cohort		
	Treated schools' mean (1)	Non-treated schools' mean (2)	Mean difference (SE) (3)	Treated schools' mean (4)	Non-treated schools' mean (5)	Mean difference (SE) (6)
A. Enrolment						
University	0.195	0.245	–0.051	0.194	0.258	–0.065
	(0.396)	(0.430)	(0.012)	(0.395)	(0.438)	(0.017)
Academic college	0.157	0.230	–0.072	0.194	0.227	–0.033
	(0.364)	(0.421)	(0.011)	(0.395)	(0.419)	(0.016)
B. Years of schooling						
University	0.788	1.053	–0.264	0.753	1.087	–0.334
	(1.897)	(2.145)	(0.058)	(1.773)	(2.153)	(0.081)
Academic college	0.495	0.810	–0.315	0.668	0.797	–0.130
	(1.332)	(1.694)	(0.045)	(1.575)	(1.655)	(0.064)
C. Labor market outcomes						
Employed (1 = Yes, 0 = No)	0.870	0.843	0.027	0.842	0.845	–0.002
	(0.336)	(0.364)	(0.010)	(0.365)	(0.362)	(0.014)
Months worked	9.319	9.067	0.252	9.018	9.090	–0.072
	(4.353)	(4.553)	(0.124)	(4.525)	(4.522)	(0.176)
Average annual earnings (NIS)	75,789	79,669	–3,880	76,821	79,483	–2,662
	(68,884)	(74,040)	(2,005)	(68,705)	(73,922)	(2,839)

TABLE 1. Continued.

	Pre: 92 and 93 cohorts			Post: 94 cohort		
	Treated schools' mean (1)	Non-treated schools' mean (2)	Mean difference (SE) (3)	Treated schools' mean (4)	Non-treated schools' mean (5)	Mean difference (SE) (6)
D. Personal status outcomes						
Married (1 = Yes, 0 = No)	0.523	0.505	0.019	0.458	0.393	0.065
	(0.500)	(0.500)	(0.014)	(0.499)	(0.488)	(0.019)
Age of first marriage	25.588	25.896	–0.309	24.982	25.269	–0.288
	(3.011)	(2.892)	(0.107)	(2.507)	(2.570)	(0.145)
Children (1 = Yes, 0 = No)	0.679	0.666	0.013	0.651	0.594	0.057
	(0.467)	(0.472)	(0.013)	(0.477)	(0.491)	(0.019)
Age of first child	26.600	27.015	–0.415	25.986	25.967	0.019
	(2.936)	(2.892)	(0.121)	(2.560)	(2.646)	(0.172)
Number of children	1.528	1.448	0.080	1.422	1.183	0.238
	(1.350)	(1.331)	(0.036)	(1.342)	(1.236)	(0.049)
E. Parental earnings						
Average father's earnings in 2000–2002	96,789	121,961	–25,172	98,271	122,977	–24,706
	(149,845)	(158,697)	(4,397)	(100,794)	(159,067)	(6,042)
Average mother's earnings in 2000–2002	49,184	59,628	–10,444	53,832	61,988	–8,156
	(65,013)	(73,989)	(1,996)	(83,998)	(73,361)	(2,932)
Average family earnings in 2000–2002	146,370	182,009	–35,639	153,005	185,386	–32,382
	(180,557)	(192,094)	(5,337)	(139,251)	(187,328)	(7,251)
Number of observations	1,565	9,157		785	4,303	

Notes: The table reports means and SDs for different post-secondary education and employment variables for eleven years after high school graduation. Each column represents these statistics for a different group as described in each column's headline. Panel A comprises binary variables indicating whether the individual was ever enrolled eleven years after high school graduation in a specific type of post-secondary institution. The categories are not mutually exclusive and overlapping is possible. Panel B reports the number of years of education an individual has attained by eleven years after high school graduation in each type of the post-secondary institutions listed in panel A. Panel C reports the mean of an employment indicator, annual earnings, and the number of months worked eleven years after high school graduation.

that the program led to an improvement in college-going rates and years of college completed without affecting university outcomes. These estimates are similar to what I will present in the next section based on the controlled DID estimation.

Summary statistics for the labor market outcomes are presented in panel C of Table 1. Eleven years after high school graduation, 87.0% and 84.3% of the individuals in the pre-treatment cohorts in the treatment and control groups, respectively, were employed, and the difference between the two was 2.7% (SE = 0.010). The respective rates in the post-treatment period are 84.2, 84.5, and –0.002 (SE = 0.014). Average annual earnings at baseline[20] were lower in the treated group by about NIS 3,880 ($2005), a gap consistent with post-secondary education treatment–control differences.

The summary indicators in panel D suggest that just over half of the treated sample is married by 2011; in the pre-treatment control group, this rate is 2 percentage points lower. The age of marriage in the treated group is 25.6, about a third of a year younger than in the control group, and a similar gap is observed in having the first child.

In panel E, I reported statistics on parental income in 2000–2002. This information, which was not available when studying the short- and medium-term effects of the school choice program, reveals gaps in favor of the control group. This is, of course, consistent with the other imbalances seen in Table 1: Father's income is higher in the pre-program control cohorts by NIS 25,172 and in the post-program by NIS 24,706, and these two differences are not statistically different from each other. The respective differences between the average mother's earnings are NIS –10,444 and NIS –8,156. These stable differences will be shown not to affect the treatment effect estimates of school choices when added as controls in the DID regressions.

4.3. Evidence on Pre-Trends

To check the assumption of the equality of pre-time trend in the treatment and control groups, one needs data on outcomes for several years before the year treatment started. However, these data are not available because the Ministry of Education in Israel has digitized information on high school outcomes at the school and student levels only since 1992. Therefore, I cannot estimate standard pre-program trends for the treatment and control groups. Nonetheless, I present evidence on equality of treatment–control pre-trends based on an alternative approach, using other data available. While these data do not include variables that would make an exact test of a pre-trend possible, the following analysis provides evidence that still alleviates to some extent the concerns about pre-trends. The data I use here are on the background characteristics and high-school outcomes of each cohort after 1986. I use these data to follow the same approach as used in a recent publication (Lavy 2020a) and examine whether there is systematic improvement in year fixed effects for schools attended

20. The mean earnings in this sample, NIS 75,789, is identical to mean earnings in the whole cohort, NIS 75,450.

by district 9 students before and after the 1994 reform. Using these data, I compare students at schools attended by district 9 students to all GR and Holon students.

Online Appendix Table A.2 presents the results of this analysis. Each column includes the results for a single outcome or characteristics: columns 1 through 5 include the five high school outcomes and columns 6, 7, and 8 include the three background characteristics. All eight variables are standardized to have mean zero and a unit SD.

Panel A presents each cohort's estimated fixed effect (Y_j), where the cohort is the year of attending the seventh grade. The dummy variable of the 1987 cohort is dropped, serving as a benchmark. Panel B includes a single fixed effect (D) for the six Tel-Aviv schools with a high share of district 9 students in the 92–94 cohorts. Since data on each school's education district are available only from 1992 onward, D is equal to 1 for students from these six schools regardless of their school district.[21]

Finally, panel C includes interaction between Y_j and D. Each of these seven dummy variable indicators (one for each cohort) measures how much the mean of characteristics and outcomes of students in schools with district 9 students differs from the respective means of other schools. Overall, there appears to be some variation between the cohorts in mean characteristics and outcomes without any noticeable time trend. Secondly, there are no significant differences in the cohort dummies between schools, with and without enrolment of students from district 9. These results suggest that the assumption of the equality of pre-time trends in the treatment and control groups is very plausible.

5. Empirical Evidence

The school choice program had positive and significant short- and medium-term effects on students' high school completion rate and academic achievements during high school. Across identification methods and comparison groups, the results consistently suggest school choice significantly reduced the dropout rate by 8.4% (35% decline) and increased the matriculation rate by 6.1 percentage points (25% improvement). These results are presented in Online Appendix Table A.3. These substantial effects were accompanied by an improvement in the quality of schooling. The average number of *Bagrut* credits increased by two units relative to the pre-program mean of 12 units, and the average score in all of the *Bagrut* exams was up by 6.6 points, about a 10% improvement. Other dimensions of quality improvement are the increase in *Bagrut* credits in science subjects and the increase in *Bagrut* honors level studies (up by a quarter relative to a mean of one such subject). These estimates are presented in Online Appendix Table A.3.

21. To check how sensitive are the results to this data limitation, I conducted a similar exercise while including in the sample only the students that enrolled in district 9's high schools. In these schools, enrolment was almost exclusively by students who reside in district 9. The results based on this sample are qualitatively the same.

The estimates based on the GD sample are presented in Table A4. They show similar positive effects of the school choice program on high school outcomes. However, conclusions about whether free school choice improves real human capital accumulation and well-being can only be based on long-term effects, particularly post-secondary enrolment and completed years of tertiary education, employment, earnings, welfare dependency, and so forth, and social outcomes to which we turn next.

5.1. Effect on Post-Secondary Schooling Attainment

I first present graphs illustrating the effect of the school choice program on post-secondary education. I focus on the two sub-sectors of academic post-secondary education in Israel. The first includes the seven research universities in Israel that confer BA, MA, and Ph.D. degrees. These universities require a matriculation diploma for admission, including an intermediate or advanced matriculation unit in English[22] and at least one matriculation subject at an advanced level. About 35% of all students are enrolled in one of the seven universities. The second sub-sector comprises more than 50 academic colleges that mostly confer a BA degree and predominantly offer social sciences, business, and law degrees.

Figure 1 presents the dynamics of the treatment effect on academic college enrolment (vertical axis) starting from the first year after high school graduation until 12 years later (horizontal axis). We note again that during the first few years after high school graduation, almost all boys (for three years) and most girls (for two years) are still in military service. Therefore, the treatment effect is not informative because it is based on a limited sample of those not drafted to service. The treatment effect is positive and statistically significant from the fourth year after graduating high school, reaching a high of 0.9% and declining slightly toward the end of the period to 5%. The respective mean enrolment rate for the treated group increases gradually from the first year and is highest at just over 16% twelve years later. The effect size here, therefore, experiences a 30% increase. The effect on completed years of academic college (presented in Figure 1(a)) increases continuously until twelve years after graduating high school, reaching a peak at 0.22 years and leveling after that. The mean of completed years of academic college in the treatment group is 0.54, and, therefore, the effect size undergoes a 40% increase.

Figures 2 and 2(a) present the estimated effects on university enrolment and attainment, and the pattern revealed in these figures is very different, as the effect is practically zero. The treated group mean of university enrolment rate is 0.19, and the mean years of university is 0.80 years, and both of these outcomes remain unchanged due to the program.

In Figures 3 and 3(a), I replicated this graphical analysis for any type of post-secondary education (including teachers' colleges and non-academic education).

22. To qualify for a matriculation-diploma, a basic study program in English is sufficient, but university admission requires a higher level.

TABLE 2. Effect of school choice on post-secondary schooling, 12 years since high school graduation.

	Enrolment		Years of schooling	
	Mean, 1992–1993 cohorts in treated schools (1)	Treatment (2)	Mean, 1992–1993 cohorts in treated schools (3)	Treatment (4)
A. Any post-secondary schooling	0.413	0.057	1.594	0.209
	(0.493)	(0.023)	(2.419)	(0.083)
B. University schooling	0.194	0.008	0.801	0.031
	(0.395)	(0.012)	(1.957)	(0.066)
C. College schooling	0.156	0.049	0.514	0.220
	(0.363)	(0.012)	(1.384)	(0.051)
Number of observations	1,555	15,268	1,555	15,268

Notes: This table presents the DID estimates of the effect of the school choice program on post-secondary schooling. Columns 1 and 2 measure enrolment into different types of post-secondary institutions, while columns 3 and 4 measure completed years of post-secondary education by institution type. The results are for 12 years after high school graduation. The variable "Any post-secondary education" refers to all different post-secondary institutions. Columns 1 and 3 .represent the mean and SD for the 1992–1993 (untreated) cohorts in the treated schools. Columns 2 and 4 report the DID estimates for each of the dependent variables. Standard errors are clustered at the school level.

The results are very similar to those presented in Figures 1 and 1(a). Overall, post-secondary enrolment increased by almost 5.7 percentage points, and total education increased by a fifth of a year.[23]

Table 2 presents detailed estimates from regressions of the effect of free school choice on post-secondary education attainment when outcomes are measured twelve years after high school graduation. The first row presents DID estimated effect on enrolment in any type of post-secondary education (column 2) and on the respective completed years of schooling (column 4). Standard errors appear below each estimate in brackets and are clustered by a secondary school. School choice enrolment in any post-secondary education increased by 5.7 percentage points relative to a pre-program mean of 41.3% in treated schools and 52.9% in the control group. The effect on completed years of education (column 4) is 0.209 (SE = 0.083). Relative to the pre-choice treatment group mean (1.594), this is a 13% gain.

It is interesting and important to know what types of post-secondary education are affected. Since the treated population is from a low socioeconomic background with a relatively low enrolment at the higher quality end of academic institutions, we expect the effect to be low on university education and higher on colleges and

23. It is important to note here that the expansion in enrollment in academic colleges due to the school choice program could not have been at the expense of other students because the choice program and the number of students affected were very small relative to the overall enrollment in academic colleges in the whole country.

non-academic post-secondary institutions. In the second row, I present the estimated effect on university education, and in the third row, I present the effect on education in academic colleges. The effect on university enrolment is practically zero (0.008, SE = 0.012), and so is the effect on university years of schooling. However, the effect on academic college enrolment is 4.9 percentage points, significantly different from zero ($t = 4.1$), and completed years of this type of education increased by over a fifth of a year (0.222, SE = 0.051). The gain in academic college education is almost equal to the overall improvement in any type of post-secondary education, indicating that the effect on any other kind of education is very small or zero.[24,25]

The evidence presented previously clearly demonstrates that the gain in post-secondary education is concentrated in the lower end of academic education in Israel. The academic colleges are less prestigious than universities, and their admission requirements are less demanding in *Bagrut* results. This pattern is perhaps expected because the treated population is mostly from a disadvantaged segment of the Israeli population. Their enrolment and years of study in a university are much lower than the country's overall mean. We can safely claim that the affected students are at the margin of being admitted to post-secondary education, which can also explain why the treatment effect is concentrated at the lower end of the quality distribution of academic education in Israel.

The long-term data that I use in this paper include information on parental income during the experiment years. It allows me to assess how sensitive the post-secondary treatment estimates are to adding controls for family earnings. The estimates presented in Online Appendix Table A.6 show that, adding to the DID regression, a control for family income (average in 2000–2002) does not change at all the point estimates relative to those presented in Table 2. For example, the estimated effect on college enrolment in Online Appendix Table A.6 is 0.047, and on college years of education, it is 0.175, almost identical to the respective estimates in Table 2. This is a remarkable result given that the treatment and control samples are not balanced in family income, but they are equally imbalanced in this dimension, as in others, for the pre-and post-treatment cohorts.

To assess how sensitive the results are to pooling the three comparison groups as one control group, Online Appendix Table A.7 presents estimates based on using as control each of the three comparison groups separately. The results are similar to those estimated using the combined control groups. For example, the estimated effect on college enrolment in Table 2 is 4 percentage points. This estimate is 5.0 percentage points when

24. Indeed, enrolment in teachers' colleges also increased by 2.7% and is significantly different from zero, but the respective increase in years of this type of education is positive but small and imprecisely measured. Enrolment and years of schooling in vocational education (two-year colleges that confer practical engineering degrees) declined by 1.5 percentage points and by 0.025 years, very small and imprecisely measured changes. These results are not presented in this paper and are available from the author.

25. In Online Appendix Table A.5, I present results of placebo effects based on a contrast between the two untreated cohorts of 1992 and 1993. Overall, these controlled experiment estimates are not significantly different from zero, similar to respective placebo regression estimates with high school outcomes as the dependent variables (Lavy 2010).

using only GR as a control group, 4.7 percentage points when using Holon as a control group, and 3.9 percentage points when using Tel-Aviv's districts 6–8. Note that the average of these three estimates is precisely 4.5 percentage points, almost identical to the treatment effect (4.9) when using the three control groups jointly. Naturally, when using each of the control groups separately, the sample is smaller, and, therefore, the estimates are less precise than when pooling all three control groups together. To further illustrate that the estimates are similar, I computed the treatment effect ratio estimated from each control group separately to the treatment effect estimated while using all control groups. These ratios are presented in columns 4, 6, and 8 of Online Appendix Table A.7. These ratios are close to 1 for almost all outcomes, especially for outcomes for which the treatment estimates are statistically significant when using the combined control groups.

Evidence Based on the GD Sample. To further check the robustness of the evidence presented previously, I use the GD design described in the previous section. The GD sample includes observations within a relatively narrow band around the municipal border between district 9 and GR. It is important to note that the treated group is from a much higher socioeconomic status in this sample, and it resembles the control group more closely. The outcomes' means of the treatment and control groups in this GD sample are presented in Online Appendix Table A.8. The means of post-secondary educational and labor market outcomes of the treated group are higher than those of district 9. In Online Appendix Table A.9, I report estimates derived from the GD sample. These estimates are similar to the estimates obtained from the full sample. The effect on academic college enrolment is 0.055, and on academic college years of education, it is 0.270. The impacts on university enrolment and years of schooling are not statistically different from zero.

5.2. *Effect on Employment and Earnings*

We start here with a graphical presentation of the impact of the school choice program on employment and earnings. Again, we measure these two outcomes for each individual based on the number of years since graduating high school. The employment and earnings data are available until 2014, so 13 years is the most prolonged period after graduating high school. We examine the effect of the program. Figure 4 presents the yearly estimates on employment. As noted earlier, the estimates for the first three years are not meaningful because most of the students in our sample were still in military service. In the fourth year after high school graduation, about 85% of the individuals in the sample were employed (according to our definition of employment, which is being employed at least for one month during the year and had positive earnings). For almost all years, the estimates are measured imprecisely: for five to seven years after high school graduation, the estimates are positive, and after that they are negative, but in most years, they are not statistically different from zero, especially in 12 and 13 years after high school graduation. Similar evidence is obtained for the outcome that measures the number of months per year of being gainfully employed.

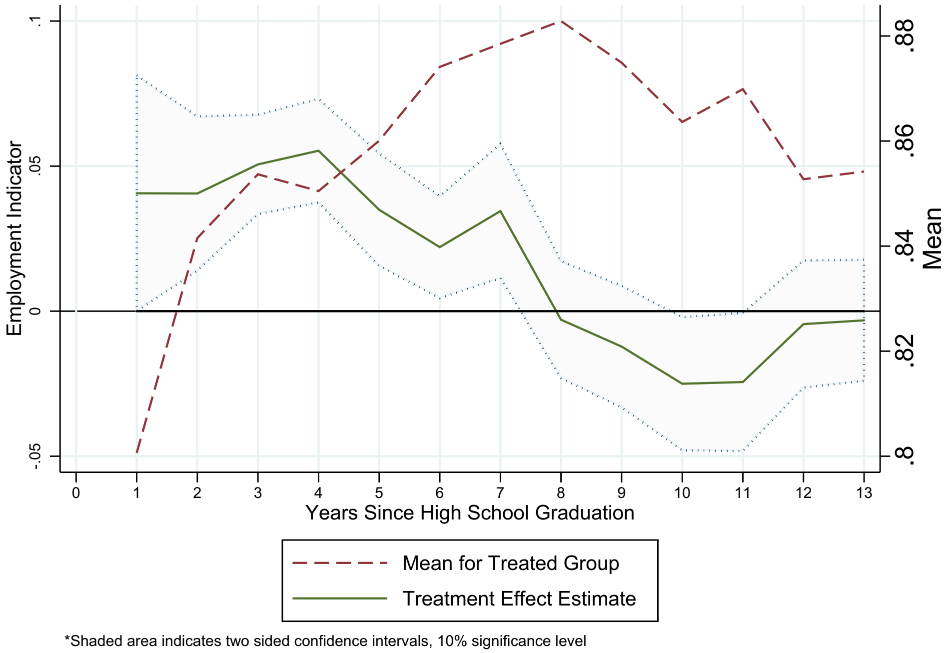


FIGURE 4. Mean and treatment effect: employment.

These imprecise and inconclusive employment dynamics imply that they do not play an essential role in determining earnings change due to the school choice program.

We next turn to the time series of estimated effects on annual earnings, presented in Figure 5. The treatment effect estimates on earnings are positive throughout the period after high school graduation. They increase over time monotonically except a spike in treatment effect nine years after high school graduation. Similarly, average annual earnings in the sample also increase monotonically until the end of the period, from NIS 40,000 (about $10,000) five years after high school graduation to just over NIS 80,000 13 years after high school graduation. The treatment effect on earnings in the last two periods of analysis is over NIS 7,000 a year. It is significantly different from zero at the 5% level of significance in both periods. Interestingly, the earnings effect is not negative even when the post-secondary enrolment rate is higher among treated students.

The pattern discussed previously is different from that reported in Lavy (2020b). Teacher pay for performance increased university education while lowering employment and earnings during the three to five years when students are studying. This contrasting pattern is not surprising for several reasons. First, it is typical that students in academic colleges in Israel work part time or full time. At the same time, university studies have a more demanding academic schedule and requirements that correspond to full total time attendance, making it more challenging to combine work with study. Secondly, the treated students in academic colleges are usually from a lower socioeconomic background than students in universities, and, therefore, they can rely

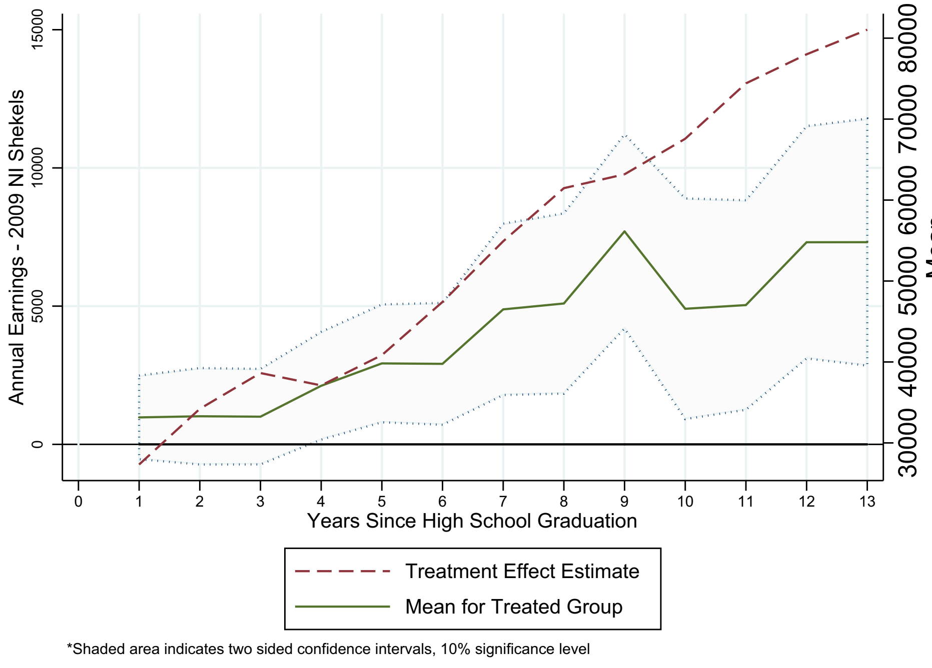


FIGURE 5. Mean and treatment effect: annual earnings—2009 NIS.

less on parental support. Thirdly, scholarships based on academic merit and low family income are available for university students but not for students in academic colleges.

Table 3 presents evidence about the effect of the school choice program on employment, number of months worked in a year, and monthly earnings, for 11–13 years after high school graduation (columns 2, 4, and 6), and based on stacking the three periods in one sample (columns 8). I also present a treatment effect on percentile rank of earnings and log earnings (practically dropping from the sample observations with zero earnings). The average employment rate in the treatment group in these three periods is 87.0, 85.3, and 85.4, respectively. The respective treatment effect on employment is negative, though it is small, and in the last two periods, it is practically almost zero. The estimated effect on months worked per year has the same pattern. It is important to note that the negative though small employment effect does not reflect a higher rate of individuals still studying among treated students; the proportion of students in the 1994 cohort who are not yet employed and are still studying is the same in both groups: in the treatment group in 2012, it is 1.4%, and in the control group, it is 1.3%. The respective means in the 1992 and 1993 cohorts are 0.7% and 0.8%, respectively.

The average annual earnings for the 1992–1993 cohorts in treated schools eleven years after high school graduation is NIS 74,412 ($17,717), 12 years after high school graduation, it is NIS 78,004 ($19,118), and 13 years after high school graduation, it is NIS 81,044 ($21,277). The school choice program's estimated effect on annual earnings is NIS 5,038 ($1,199) eleven years after high school graduation, NIS 7,311 12

TABLE 3. Effects of school choice on employment and earnings, by years since high school graduation.

	11 years		12 years		13 years		Stacked regression 11–13 years	
	Mean, 1992–1993 cohorts in treated schools (1)	Estimate (2)	Mean, 1992–1993 cohorts in treated schools (3)	Estimate (4)	Mean, 1992–1993 cohorts in treated schools (5)	Estimate (6)	Mean, 1992–1993 cohorts in treated schools (7)	Estimate (8)
Employment indicator (1 = Yes, 0 = No)	0.870	–0.024	0.853	–0.004	0.854	–0.003	0.858	–0.009
	(0.337)	(0.014)	(0.354)	(0.013)	(0.353)	(0.013)	(0.349)	(0.011)
Total annual earnings (2009 NI sherels)	74,412	5,038	78,004	7,311	81,044	7,314	77,926	6,367
	(64,417)	(2,304)	(67,284)	(2,555)	(70,190)	(2,716)	(67,592)	(2,319)
Months worked	9.310	–0.235	9.246	-0.191	9.263	–0.042	9.256	–0.143
	(4.357)	(0.193)	(4.460)	(0.155)	(4.437)	(0.155)	(4.429)	(0.145)
Percentile rank	41.260	2.208	40.602	2.360	40.339	2.682	40.781	2.368
	(30.572)	(1.188)	(30.563)	(1.112)	(30.467)	(1.105)	(30.594)	(1.032)
Number of observations	1,560	15,286	1,555	15,268	1,550	15,228	4,737	46,217
Log (earnings)	11.026	0.103	11.112	0.040	11.122	0.119	11.088	0.081
	(1.008)	(0.038)	(0.940)	(0.039)	(1.078)	(0.037)	(1.012)	(0.029)
Number of observations	1,359	13,172	1,333	13,148	1,330	13,123	3,959	39,709

Notes: Columns 1 and 2 report results for 11 years after high school graduation, columns 3 and 4 for 12 years, and columns 5 and 6 for 13 years. Columns 1,3, 5, and 7 report the mean and SD for the 1992–1993 (untreated) cohorts in the treated schools. Columns 2, 4, 6, and 8 report the DID estimates for each of the dependent variables listed above. Standard errors are clustered at the school level.

years after, and NIS 7,314 13 years after. All three estimates are significantly different from zero at the 5% level of significance. I also estimated an earnings effect using a combined 11 to 13 years after high school graduation earnings, stacking the data together for these three periods. This estimate is NIS 6,367 (column 8 of Table 3), close to the average (6,554) of the estimates for 11, 12, and 13 years after high school graduation.

The estimated effects on percentile rank and on log earnings are positive and statistically significant. They are more precise on the latter. The stack regression on log earnings indicates an increase of 6.8% on earnings among the employed.

The lower part of Online Appendix Table A.7 presents estimates of the effect on labor market outcomes based on using each of the three comparison groups separately as control. The results are not different from those shown in Table 3. A notable exception is the labor market outcome estimates obtained when using only Tel-Aviv's districts 6–8 as the only control group. In the combined control group sample, these estimates are large and positive for annual earnings and small and insignificant for employment (Table 3). When using only Tel-Aviv's districts 6–8 as a control group, the effect on earnings becomes insignificant, and the effect on labor supply becomes more negative. It could be that this small and insignificant effect results from noisier earnings data in districts 6–8, which are populated by many immigrants and illegal foreign workers and also a large number of the minority Arab population. These groups have a tradition of less precise earnings records at the tax authorities. Nevertheless, it is essential to note that the estimates obtained when GR or Holon is used as a control group are very similar.

Online Appendix Table A.10 presents the treatment estimates on earnings and employment when controls are added for parental or family earnings. The three columns correspond to estimates for 11, 12, and 13 years after high school graduation. These estimates are very similar to those presented in Table 3, implying that adding control for parental earnings does not affect these treatment estimates.

In Online Appendix Table A.11, I present the estimated effect on labor market outcomes based on the GD sample. The impact on earnings has the same pattern as in Table 3, but the estimates are larger. Focusing on the stacked data estimates in column 8, the annual earnings' gain during 11–13 years since high school graduation is NIS 7,939—a 9.1% increase relative to total annual earnings in this period. This large effect is partly due to the higher increase in post-secondary education and the higher rate of return for university education in Israel relative to the rate of return for academic college education.

Caplan et al. (2009) report that academic college graduates earn 20%–30% less than university graduates in their first jobs in many fields of study. However, the effect on employment is negative and larger than what is observed in the full sample. This puzzling pattern is resolved when I estimated all treatment effects by gender, which shows that all of the negative adverse effects on employment are on women. Most of the positive effect on earnings is due to men. Consistent with these results is the positive effect on women's marriage rate and fertility without a corresponding effect on men. I discuss these results in more detail in later sections where I report results by gender and the marriage and fertility effect of school choice.

It is useful to compare the evidence presented previously to the evidence presented in Wondratschek, Edmark, and Frolich (2013), the only other school choice study that examined the long-term effect on several outcomes of college education and employment. It evaluates the effects of a large-scale school choice reform in Sweden implemented in 1992, which significantly increased the amount of choice in compulsory schooling among the already existing public schools and private schools that were mostly opened later. This study's results reveal very small effects on the short-term outcomes (test scores and grades) and zero effects on longer-term outcomes (employment, higher education, criminal activity, and health). As an explanation for these findings, the authors suggest that the previously existing Tiebout choice (i.e. moving homes) may already have provided sufficient choice options for those families who wanted to choose. Another explanation could be that the long-term outcomes are measured at a relatively young age: crime and health at age 22 and university degree and employment at age 25. The long-term effects that I present in this paper are measured at a much older age. Even when estimated at age 22–25, they are very small and imprecise.

Comparing the Effect on Earnings to Related Evidence. This is the first study to provide evidence of school choice's effect on students' earnings in adulthood. However, it is still useful to compare our results to other childhood and education interventions on adulthood earnings. Andersson et al. (2016) estimated that living during teenagehood in public or voucher housing increased females' earnings by 18%–21%. Each additional year of public or voucher-supported housing increases earnings by 7% for females. For males, each year of public housing participation as a teenager increases adult earnings by 5% with no corresponding effect of voucher housing. Chetty et al. (2011) have shown that having a kindergarten teacher with more than ten years of experience increased students' average annual earnings at ages 25–27 by 6.9% ($1,093) between 2005 and 2007. Similarly, an improvement in class quality increased the average yearly income earned between ages 25 and 27. Johnson et al. (2016) show that for children from low-income families, increasing per-pupil spending by 10% in all twelve school-age years increased adult hourly wages by 13%.

Schweinhart et al. studied the long-term effect of the High/Scope Perry Preschool experiment. They found that students in the treatment had significantly higher median annual earnings than the no-program group: 20% higher at age 27 and 36% higher at age 40. Finally, Chetty, Hendren, and Katz (2016) find that moving to a lower-poverty neighborhood (Moving to Opportunity [MTO]) significantly improves college attendance rates—by 2.5%—and earnings by 31% for young children (below age 13) when their families moved. Our estimated effects on earnings are not unusually high relative to estimates surveyed previously. For example, the teachers' pay experiment raised college enrolment by 5%, twice that of the MTO effect, and increased earnings 10–12 years after high school graduation by 7%–9%, a quarter of the MTO effect.

5.3. Does the Change in School Structure Confound the Estimated Effect of School Choice?

As noted in Section 2, parallel to the school choice reform in district 9, the Tel-Aviv school authority changed the structure of four high schools in the city, from three-grade schools to six-grade schools, by merging into high schools all middle schools. All of the city's other post-primary schools were already structured this way. However, it is important to check whether the results I reported previously are potentially confounded by the structural change in these four schools.[26] The evidence presented in this section reassures that this is very unlikely.

First, Holon's education system, one of the three control groups that I use in this paper, has also moved to a six-grade structure around the same time that Tel-Aviv did. Holon made the change in 1995 and Tel-Aviv in 1994.[27] This implies that Holon students were exposed to the new six-grade school structure from the eighth grade, while the treated students in Tel-Aviv were exposed to the same change from the seventh grade. Thus, Holon students had five years of exposure to the new school structure, while district 9 students had six years of exposure. This minor difference is unlikely to matter for the medium-term effect at the end of high school and for the longer-term effect presented in this paper. Indeed, as shown previously, the school choice effects on medium- and long-term outcomes estimated based on using Holon as a control group are identical to the estimates when using GR as a control group.

Secondly, as noted in Section 2, school districts 6–8 in Tel-Aviv (one of the three control groups I use for identification) implemented the change in school structure at the same time that district 9 did. As shown previously, the estimated effects on high school and post-secondary schooling based on using districts 6–8 as a control group are similar to those obtained when using GR (or Holon) as a control group. The only difference is the estimated effect on earnings, which I discuss in Section 5.2.

5.4. *Effect of School Choice by Gender*

Previous research on the effectiveness of schooling interventions has shown differences in the responsiveness of boys relative to girls (e.g. Angrist and Lavy 2009). To test for this possibility, I stratify the sample based on students' gender and present in Table 4 the effect of school choice on boys and girls' high school outcomes separately.[28] Free school choice improved all five outcomes for boys and girls except girls' dropout rate, which declined by only 2.1 percentage points, with a *t*-statistic of

26. The school structure reform that incorporated the three middle school grades in high schools was part of a national program that was implemented gradually in all school districts in the country. This reform kept key features of middle schools unchanged. which likely reduced the scope for potential confounding effects. For example, middle schools remained an independent entity, having an autonomous principal, unchanged educational curriculum, and a separate budget. Its teachers continued to get their education in teachers' colleges (while high school teachers had to obtain university degrees), and they remained organized in the same separate union, working under a separate collective bargaining agreement with wages lower than those of high school teachers.

27. Evidence regarding the year of reform in Holon is available from websites of schools that existed at the time. For example, in the front page of the Sharet secondary school link: https://www.holon.muni.il/Residents/Education/AlYesody/Lists/List/CustomDispForm.aspx?ID=32. Similarly, for Katzir secondary school: http://www.katzir.hs.holonedu.org.il/BRPortal/br/P102.jsp?ar. All high schools in the city made the change in the same year.

28. These results are not reported in Lavy (2010).

TABLE 4. Effect of school choice on high school outcomes, by gender.

	Boys		Girls	
	Mean, 1992–1993 cohorts in treated schools (1)	Treatment (2)	Mean, 1992–1993 cohorts in treated schools (5)	Treatment (6)
Drop-out	0.266 (0.442)	–0.101 (0.027)	0.103 (0.304)	–0.021 (0.014)
Eligible for Bagrut	0.377 (0.485)	0.071 (0.024)	0.513 (0.500)	0.090 (0.029)
Average score	53.499 (37.166)	8.856 (1.784)	66.823 (32.578)	3.968 (1.643)
Number of science credits	1.414 (3.101)	0.466 (0.199)	1.709 (3.606)	0.333 (0.201)
Number of credit units in matriculation exams	14.427 (11.641)	2.913 (0.590)	17.312 (10.000)	1.167 (0.541)
Number of honor-level subjects	1.390 (1.540)	0.386 (0.085)	1.698 (1.346)	0.182 (0.068)
Number of observations	801	8,093	741	7,601

Notes: This table presents the DID estimates of the effect of the school choice program on high school outcomes for boys and girls separately. Columns 1 and 3 represent the mean and SD for the 1992–1993 (untreated) cohorts in the treated schools. Columns 2 and 4 report the DID estimates for each of the dependent variables. Standard errors are clustered at the school level.

just 1.5. However, it is noticeable that girls have higher means in all six high school outcomes and girls' dropout rate is much lower than that of boys, 10.3% versus 26.6% among boys in the sample. However, all of the point estimates suggest that the effects are larger among boys except for the matriculation rate. The gender differences are statistically significant for the following outcomes: dropout rate, average score, number of credits, and number of subjects studied at honors classes.

In Table 5, I report the estimated effects of school choice on post-secondary schooling outcomes by gender. Boys gained an 8.9 percentage increase in enrolment and a fourth of a year of post-secondary schooling, all of it in academic colleges. The girls' gain is also limited to academic colleges, but the estimated effect is much smaller than that for boys, 3 percentage increase in enrolment, and a 0.13 year of schooling.

Table 6 shows that these gender differences are also reflected in the labor market. Focusing on the estimates based on the stacked regressions that pool data for 11–13 years after high school graduation, the effect on boys' earnings is NIS 7,711 (SE = 3,691), an 8% increase, while the earnings effect for girls is NIS 4,080 (SE = 2,240). However, the girls' labor market experience includes a 2.3% decline in employment rate and over half a month decline in months of work in a given year. Both of these effects are precisely measured. The decline in months of work among treated girls accounts for a negative NIS 3,700 (= [(67,963/9.330) × 0.533)]) effect, which reduces the expected

TABLE 5. Effect of school choice on post-secondary schooling, 12 years since high school graduation.

	Boys				Girls			
	Enrolment		Years of schooling		Enrolment		Years of schooling	
	Mean, 1992–1993 cohorts in treated schools (1)	Estimate (2)	Mean, 1992–1993 cohorts in treated schools (3)	Estimate (4)	Mean, 1992–1993 cohorts in treated schools (5)	Estimate (6)	Mean, 1992–1993 cohorts in treated schools (7)	Estimate (8)
A. Any post-secondary schooling	0.360	0.089	1.376	0.257	0.469	0.023	1.827	0.148
	(0.480)	(0.032)	(2.281)	(0.111)	(0.499)	(0.028)	(2.540)	(0.135)
B. University schooling	0.155	0.006	0.688	–0.018	0.235	0.007	0.921	0.076
	(0.362)	(0.019)	(1.901)	(0.111)	(0.424)	(0.024)	(2.010)	(0.130)
C. College schooling	0.144	0.068	0.456	0.302	0.169	0.030	0.577	0.130
	(0.351)	(0.020)	(1.255)	(0.073)	(0.375)	(0.015)	(1.510)	(0.062)
Number of observations	805	7,855	805	7,855	750	7,413	750	7,413

Notes: This table presents the DID estimates of the effect of the school choice program on post-secondary schooling for boys and girls separately. Columns 1 and 2 and 5 and 6 measure enrolment into different types of post-secondary institutions, while columns 3 and 4 and 7 and 8 measure completed years of post-secondary education by institution type. The variable "Any post-secondary education" refers to all different post-secondary institutions. Columns 1, 3, 5, and 7 represent the mean and SD for the 1992–1993 (untreated) cohorts in the treated schools. Columns 2, 4, 6, and 8 report the DID estimates for each of the dependent variables. Standard errors are clustered at the school level.

TABLE 6. Effect of school choice on employment and income, by gender and years since high school graduation.

	11 years		12 years		13 years		Stacked regression 11-13 years	
	Mean, 1992–1993 cohorts in treated schools (1)	Estimate (2)	Mean, 1992–1993 cohorts in treated schools (3)	Estimate (4)	Mean, 1992–1993 cohorts in treated schools (5)	Estimate (6)	1992–1993 cohorts in treated schools (7)	Estimate (8)
A. Boys								
Employment Indicator (1 = Yes, 0 = No)	0.853	–0.003	0.837	0.006	0.839	0.011	0.843	0.005
	(0.354)	(0.023)	(0.369)	(0.022)	(0.368)	(0.022)	(0.364)	(0.020)
Total annual earnings (2009 NIS)	82,485	5,668	86,377	8,895	91,666	9,807	87,225	7,711
	(72,969)	(3,683)	(76,273)	(3,975)	(78,715)	(4,459)	(76,434)	(36,181)
Months worked	9.220	0.231	9.178	0.099	9.185	0.360	9.189	0.246
	(4.540)	(0.292)	(4.627)	(0.251)	(4.618)	(0.262)	(4.584)	(0.246)
Number of observations	810	7,867	805	7,855	802	7,819	2,480	23,706
B. Girls								
Employment indicator (1 = Yes, 0 = No)	0.888	–0.046	0.869	–0.013	0.870	–0.018	0.874	–0.023
	(0.316)	(0.017)	(0.337)	(0.019)	(0.336)	(0.020)	(0.332)	(0.014)
Total annual earnings (2009 NIS)	65,692	3,595	69,018	4,738	69,655	3,879	67,963	4,080
	(52,339)	(2,944)	(54,697)	(2,379)	(57,628)	(2,708)	(54,904)	(2,240)
Months worked	9.408	–0.714	9.319	–0.463	9.346	–0.466	9.330	–0.533
	(4.151)	(0.249)	(4.275)	(0.233)	(4.237)	(0.230)	(4.245)	(0.183)
Number of observations	750	7,419	750	7,413	748	7,409	2,287	22,511

Notes: This table presents the DID estimates of the effect of the school choice program on different employment and earnings outcomes for boys and girls separately. Columns 1 and 2 report results for 11 years after high school graduation, columns 3 and 4 report results for 12 years, and columns 5 and 6 report results for 13 years. The variable "Employment indicator" equals 1 if an individual has any work record for the given year and 0 otherwise. Columns 1,3, 5, and 7 report the mean and SD for the 1992–1993 (untreated) cohorts in the treated schools. Columns 2, 4, 6, and 8 report the DID estimates for each of the dependent variables listed above. Standard errors are clustered at the school level.

positive effect of the increase in college education on earnings. This reduction is almost equal to the gender difference in the effect of school choice on earnings.

On the other hand, the effect on boys' employment is not statistically significant. What can explain the different patterns of the labor market effects by gender, particularly the negative effect on girls' employment? In Section 5.5, I present the effect of school choice on marriage and fertility outcomes and use them to answer this puzzle partially.

5.5. Effect on Marriage and Children

In Table 7, I report results regarding the program's effect on marriage and fertility. The marriage and fertility data are available up to 2011. However, I can still compute these outcomes by years after high school graduation because the dates of marriage and children's birth are available in the data. A total of 52% of the treated sample are married after ten years following high school graduation. The treatment effect on being married at this date is 3.3%, but it is not precisely measured. The age of marriage effect is 0.218 (SE = 0.146). The estimated impacts on the indicator of having children and on the number of children are positive, but they are not measured precisely. However, the age of having the first child is delayed by two-thirds of a year, and this effect is precisely measured with SE = 0.177.

In columns 4 and 6, I present the respective effects on boys and girls, and meaningful differences are revealed here. School choice has a negative though imprecise effect on boys' marriage rate, and it has a large positive effect on girls' marriage rate, which is 8.9% higher relative to a mean of 57.4%. The age of girls' first marriage is unchanged, while it is delayed by over a third of a year for boys. The probability of having children by age 28 is increased for girls by 7.7%, relative to a mean of 72.7%, while the boys' effect is zero. The effect on the number of children is positive for girls, up by 0.238 children, while it is zero for boys. All the estimated effects on girls are statistically significant. The higher marriage and fertility rates among women in the treated sample could be related to the heterogeneity by gender in the treatment effect on labor market outcomes, which we reported in the previous section. The lower treatment effect on women's employment and earnings contrasts to the similar or even higher treatment effect among women on post-secondary schooling outcomes relative to untreated women in the sample.

The first point to note in this analysis is the evidence in Table 2, which shows that the school choice program's effect on various high school outcomes is higher for girls by gender. Unlike the results documented in Deming et al. (2014), the initial gender gap in achievement is much smaller and hardly significant, with both genders displaying positive, significant treatment effects. The particularly important effect on eligibility for matriculation is even higher among girls. The lower effects for girls in these groups could be attributed to higher outcomes before the reform. The first significant gap between effects for boys and girls appears in post-secondary schooling.

In contrast to Deming et al. (2014), boys exhibit a positive (albeit imprecise) effect in university enrolment and years of schooling. Girls reveal no such rise. However,

TABLE 7. Effect of school choice on marriage and fertility, 10 years after high school.

	Entire sample		Boys		Girls	
	Mean, 1992–1993 cohorts in treated schools (1)	Estimate (2)	Mean, 1992–1993 cohorts in treated schools (3)	Estimate (4)	Mean, 1992–1993 cohorts in treated schools (5)	Estimate (6)
Children (1 = Yes, 0 = No)	0.679	0.037	0.634	0.001	0.727	0.077
	(0.467)	(0.022)	(0.482)	(0.032)	(0.446)	(0.025)
Number of children	1.528	0.118	1.362	–0.004	1.707	0.238
	(1.350)	(0.062)	(1.324)	(0.075)	(1.355)	(0.076)
Age of first child	26.600	0.626	27.316	0.514	26.067	0.640
	(2.936)	(0.177)	(2.530)	(0.263)	(3.103)	(0.238)
Married (1 = Yes, 0 = No)	0.523	0.033	0.477	–0.020	0.574	0.089
	(0.500)	(0.023)	(0.500)	(0.031)	(0.495)	(0.030)
Age of first marriage	25.588	0.218	26.561	0.362	24.752	0.138
	(3.011)	(0.146)	(2.537)	(0.170)	(3.134)	(0.222)
Number of observations	1,542	15,227	814	7,911	751	7,423

Notes: This table presents the DID estimates of the effect of the school choice program on different personal outcomes. Columns 1 and 2 report results for the entire sample, while columns 3 and 4 and 5 and 6 report the results for boys and girls, respectivly. Columns 1, 3, and 5 report the mean and SD for the 1992–1993 (untreated) cohorts in the treated schools. Columns 2, 4, and 6 report the DID estimates for each of the dependent variables listed above. Standard errors are clustered at the school level.

this gap does not appear for college outcomes—where both genders have almost identical treatment estimates. Therefore, at this stage, the gap appears, if at all, only among those in the higher end of the education distribution. These effects speak of the potential for heterogeneous treatment effects.

The gender gap in the treatment effect on labor market outcomes is indeed striking. Women display adverse labor supply effects and much smaller effects on earnings. In contrast, men have positive and significant effects on earnings and smaller and statistically insignificant labor supply effects. The reduction in labor supply can partially explain the divergence in annual earnings—without the reduction of half of a month of work, and women would have enjoyed a treatment effect on earnings similar to that of men.

The table below shows a similar average reduction in wages among married women.

	Labor market statistics, means, and SDs among women	
	Labor supply (months)	Annual income (NIS)
Unmarried	9.039 (4.431)	76,586 (65,984)
Married	8.891 (4.410)	72,561 (63,022)
Average effect of marriage	–0.148	–4,025

The reduction in labor supply corresponds directly to an *increase* in marriage and fertility rates of 8%–9%. This is similar to the decrease in labor supply, 6% in months worked and 3% in employment. Therefore, it is likely that most of the statistically significant gender differences in the effect of school choice on labor market outcomes can be attributed to a rise in marriage and fertility rate among women, which itself can result from an improvement in educational attainment (specifically improved *Bagrut* eligibility, a prime educational outcome in Israeli society). In this regard, it is relevant to note that in Israel, most women and men are at least two to three years older than American and European counterparts when making decisions about college schooling. This age difference is due to the compulsory military service in Israel, three years for men and two years for women, commencing at the end of high school. Therefore, any gender-specific attenuation of treatment effects through marriage, at the same stage in the post-secondary schooling process, could be larger for Israelis.

6. Evidence on Mechanisms

In Lavy (2010), I provided evidence about the effects of school choice on the short- and medium-term academic outcomes. Some of it suggested that the program led to a better match between students and schools. In this paragraph, I present additional evidence about the role of this mechanism. After the free choice program, most of district 9 chose a different school from the one they otherwise would have been assigned to

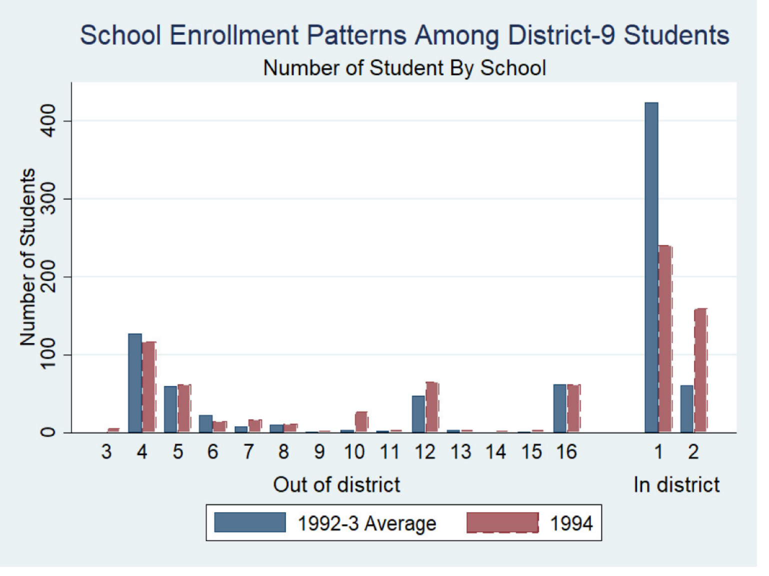


FIGURE 6. School enrollment patterns among district-9 students.

under the pre-choice assignment rules. Second, after introducing free school choice, there was a dramatic decline in students' mobility between schools at each potential junction between grades, from the seventh to twelfth grade. Third, a large proportion of district 9 students who had the longest travel distance from home to schools in districts 1–4 also chose an out-of-district school, which suggests willingness of students to "pay" in terms of longer travel time to enroll in what they perceive to be a better school.

Also shown in Lavy (2010) is that competition among schools intensified following school choice and led to improved school quality. In this section, I provide additional evidence that school quality was an important channel of the effect of school choice. Using pre-program cohorts' data, I measure two (related) aspects of school quality (similar to the approach used in Deming 2011). The first is the quality of peers, measured by peers' mean background characteristics. The second is school average academic achievements. Naturally, school choice might depend on these two aspects of school quality. Therefore, this analysis's starting point is to examine whether high-performing students opted out to better schools more than others. Figure 6 presents the school enrolment pattern of district 9 students before the school choice reform (1992–1993) and after the reform (1994) for out-of-district schools and in-district schools. School attendance of district 9 students in out-of-district schools hardly changed. However, there is a very significant shift of students from an existing (before the program) school in district 9 to a district 9 school before the program had only grades 10–12. When the program started, it expanded its grade coverage to also include grades 7 [to]9. Figure 7 shows that the proportion of district 9 students in each of the out-of-district schools also remained unchanged. The evidence in these two

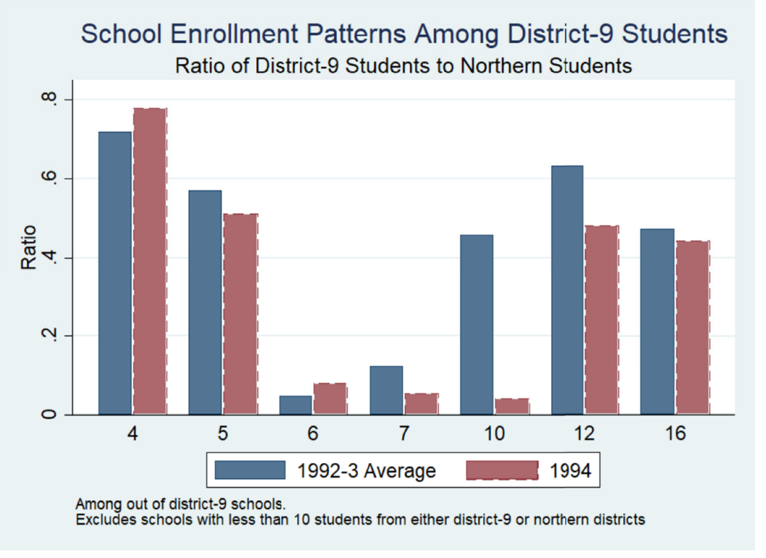


FIGURE 7. Excludes schools with more than 10 students from ether district-9 or northern districts.

figures suggests that if sorting played a role in the program's effects, it was mostly through the matching of students to schools that are more effective for them, rather than through an increase in students' enrolment in better schools.[29]

These results and the evidence presented in Figures 6 and 7 suggest that an aggregate *improvement* in the learning environment is not the result of a large-scale exodus of students to better schools in North Tel-Aviv. This implies that there was no "cream-skimming" due to the implementation of the school choice program. To check more formally the choice patterns of 1994 district 9 students, I first calculated the mean of school quality and of peer characteristics in the 1992–1993 cohort. I then computed the weighted mean of these averages, first, by using as weights the proportion of district 9 students who attended each school during the 1992–1993 cohorts and, second, by using as weights the proportion in each school of district 9 students in the 1994 cohort. I then standardize these variables to have a zero mean and a unit SD (Z-score). The sample includes 14 schools, which I use in a second step to measure the degree to which the students of the 1994 cohort improved their schooling environment

29. A 1997 report (Levy, Levy-Keren, and Liberman 1997) prepared for the Tel-Aviv municipality school authority offers more evidence for the lack of significant systematic sorting in the first year of the program. This evidence was based on standardized tests in mathematics and in Hebrew that all Tel-Aviv students took in the year following the implementation of the school choice program. It shows that the test scores' means of students from district 9 who attended in district schools and the means of students who attended out-of-district schools are not different. This study also provides evidence on lack of systematic selection between students within district 9, and concludes that the Tel-Aviv choice program did not cause an imbalance in the proportion of high-ability students between those who enrolled in in-district schools and those who enroled in out-of-district schools. The same conclusion was drawn by the city education authorities, as stated in the city council records (city council protocol from March 5, 1995).

through choice. I first stacked the 1992–1993 and 1994 means of all variables and then used this sample to estimate regression models where the dependent variable is one of these weighted means. The right-hand side variable is an indicator for the 1994 cohort. The stacked data on peer quality for each school include the means of all peer characteristics, and the stacked data on school quality include the school means of *Bagrut* outcomes. The 1994 cohort indicator's estimated coefficient captures the average change in peer quality or school quality achieved through school choice. I then tested whether the pattern of choice in 1994 was different from the two previous years' pattern of choice. I assumed throughout this analysis that students based their decision on observed characteristics of schools. The results are presented in the following table.

Comparison of high school peer quality and school mean *Bagrut* outcomes given school choice patterns of district 9 students in cohorts before and after the choice reform

	Peer quality (mean Z-Score)	School quality (mean Z-Score)
(Clustered SE)	(0.215)	(0.310)
(Robust SE)	[0.227]	[0.236]

Column 1 presents the results from a regression of the mean of peer characteristics as the dependent variable. Columns 2 presents estimates from a similar regression, where the mean of school quality (Z-score) is the dependent variable. Both estimates are positive and almost identical. The estimate in the first implies that district 9 students improved their peer quality by 0.397 SD. The second column estimate suggests that district 9 students experienced a 0.373 SD increase in their school quality. The gain in peer quality is statistically significant, but the gain in school quality is not (when SEs are clustered by school).

Based on these results, I explore the effect of school quality as a mechanism for the impact of school choice. I estimate a student-level outcome regression using a sample that includes only pre-program cohorts and include the regression student's characteristics and school fixed effects. I then extract the school fixed effects, viewing them as a measure of school quality. These school fixed effects capture each school's common factor of students' mean *Bagrut* exam test scores net of the contribution of all other covariates. I then standardize the schools' fixed effects to a zero mean and unit SD Z-score. In a second step, I add this school quality measure as a determinant in a regression of treated students' (the post-reform cohort) outcomes in the DID regressions, both as a main effect and as an interaction with the treatment indicator. In other words, the estimated parameter of this interaction term represents the degree to which school quality mediates the effect of school choice on treated students.

Beyond the possibility that there might be other mechanisms at play (e.g. better student–school match, which I discussed previously), I also note here as a word of caution that the school quality variable is not entirely exogenous because students choose their preferred secondary school. However, the analysis conducted previously has shown that there was not much selection in the choice between in-district and out-of-district schools. The quality of peers, based on both mean characteristics and outcomes, did not vary with school quality. Therefore, we can proceed with this analysis even though identifying the effect of school quality is not "safe proof". Another reason not to be too concerned about the endogeneity of schooling quality is that the overall effect of the program from the model that includes an interaction effect between school choice and school quality, evaluated at the mean of schools' quality, is very similar to the estimated treatment effects presented in Tables 2 and 5.

I present in Table 8 the estimates of the main effect of school choice and of its interaction terms with school quality. Panel A presents results for high school outcomes, panel B for post-secondary schooling, and panel C for labor market outcomes. In columns 1, 3, and 5, I present the school choice's main effect, and columns 2, 4, and 6 estimate the interaction between school choice and school quality. First, I note that the estimates of the main effect of school choice are positive for all outcomes, and in almost all cases, they are statistically different from zero. The exceptions are those variables for which the effect was also not significant when the interaction term with school quality was not included. Second, the estimated interaction term between school choice and school quality is consistently positive, suggesting that the effect of school choice increases with school quality though with varying effect size. Third, the effect of school choice rises in most cases with school quality, indicating that the positive effect of school choice is driven partly by the good schools in the sample. For example, the effect of school choice on the average score, when evaluated at about 0.5 SD above the mean of school quality, is 4.35 points. It jumps to about ten points when evaluated at the mean ofschool quality. On the other hand, moving down away from the mean of school quality quickly reduces the positive effect of school choice.

6.1. What Explains the Increase in Earnings?

Note that if we assume that all of the 8% average increase in annual earnings 11–13 years after high school graduation (based on the DID estimates using the full sample and stacked data) is due to the 0.2 increase in years of schooling, this will imply a much higher rate of return to a year of education estimated in recent studies in Israel (Frisch and Moalem 1999; Frisch 2007).[30] However, as shown previously, treated students experienced a range of improvements in educational outcomes that are likely to be rewarded in the labor market independently of the return to post-secondary years of schooling. Particularly important is the matriculation rate, which increased by 6%–7%

30. It should be noted that the sample used in this analysis includes individuals with zero earnings. Therefore, the estimated impact on earnings could also reflect an indirect effect through an effect on employment, while a classic Mincer rate of return to schooling regression does not include individuals with zero earnings. However, the evidence in Table 3 did not reveal any effect on employment.

TABLE 8. Estimates of treatment effect heterogeneity, by school quality.

A. High school outcomes			B. Post-secondary outcomes			C. Labor market outcomes		
	Main treatment effect (1)	Treatment × school quality (2)		Main treatment effect (3)	Treatment × school quality (4)		Main treatment effect (5)	Treatment × school quality (6)
Drop-out	–0.029 (0.018)	–0.127 (0.034)	Any post-secondary, Enrolment	0.047 (0.026)	0.096 (0.046)	Employment indicator	–0.011 (0.013)	0.037 (0.032)
Eligible for matriculation diploma	0.082 (0.024)	0.092 (0.034)	Any post-secondary, years	0.202 (0.091)	0.456 (0.217)	Total annual earnings	7,422 (2,786)	11,463 (6,335)
Average score	4.35 (1.608)	11.973 (3.127)	University, enrolment	0.013 (0.013)	0.025 (0.031)	Percentile rank	2.322 (1.176)	5.317 (2.319)
Matriculation units	1.762 (0.509)	3.22 (0.919)	University, years	0.090 (0.073)	0.089 (0.169)	Months worked	–0.129 (0.166)	0.470 (0.385)
Science matriculation units	0.587 (0.161)	0.265 (0.269)	College, enrolment	0.039 (0.015)	0.058 (0.029)			
Number of honor-level subjects	0.273 (0.068)	0.36 (0.121)	College, years	0.171 (0.065)	0.334 (0.102)			

Notes: Standard errors are presented in parentheses and are clustered by school.

and earned a return of about 13% independently of the return to years of education.[31] Also, the quality improvements in the matriculation study program and diploma (e.g. the average score, number of credit units, and credits in honors and science subjects) could be rewarded in the labor market beyond the return to years of schooling.[32]

The program also led to improvements in some non-cognitive behavioral outcomes. For example, it reduced violence and classroom disruption, improved teacher–student relationships, and increased students' social acclimation and satisfaction at school. These effects suggest that the program improved students' social skills. Recent evidence indicates that the labor market increasingly rewards such skills.[33]

The interesting question is whether the gains experienced by students due to the access to free school choice (particularly the increase in academic college entry, completed years of education, and higher earnings in adulthood) could be predicted by the short- and medium-term positive effects of school choice on *Bagrut* outcomes? That is, are the effects measured at the time of the experiment predictive of the program's long-term effects? Do *Bagrut* outcomes that measure the quality of study program play an equal role in this regard? For example, a matriculation certificate is a prerequisite in virtually all post-secondary programs in Israel. A matriculation certificate is a requirement in many early-career jobs that do not require a college degree. Thus, the matriculation channel of the program's effect on labor market is twofold. Intuitively, to decompose the effect, we need to know the causal effect of matriculation on post-secondary outcomes, the labor market returns to a matriculation certificate, and the labor market returns to the different types of post-secondary schooling. Unfortunately, our research design does not allow getting unbiased estimates for these returns.[34] Therefore, given the endogeneity threat, I can only provide some suggestive and correlative evidence on each mechanism's importance. I use Sobel–Goodman mediation test's logic to examine whether getting a matriculation certificate is an essential pathway for the program's influence on long-term post-secondary and labor market outcomes.[35] If this is indeed the case, we should observe a reduction in the coefficient of the program when adding high school matriculation outcomes as a control in the regression. A similar logic follows when estimating the mediation effects of post-secondary education on labor outcomes. A recent application of this procedure is demonstrated in Satyanath, Voigtlaender, and Voth (2017).

31. For example, Angrist and Lavy (2009) suggest that *Bagrut* holders earn 13% more than other individuals with exactly twelve years of schooling, but this estimate does not account for selection by ability in obtaining matriculation.

32. Caplan et al. (2009) demonstrate that earnings in Israel are highly positively correlated with the quality of post-secondary schooling (colleges versus universities and higher versus lower quality universities). For example, this study shows that earnings are much higher for graduates of Tel-Aviv, Jerusalem, and the Technion Universities relative to graduates from the other four universities in the country. Admission to the top universities is of course positively correlated with the high school matriculation outcomes.

33. See. for example. Deming (2015).

34. I am also unaware of any reliable studies on the causal direct effect of receiving a matriculation degree on labor market outcomes in Israel.

35. See Baron-Kenny (1986) for a canonical reference of the Sobel–Goodman test.

Following this approach, I first estimate ordinary least-squares regressions of annual earnings on the various high school educational outcomes, including in the regression controls for students' parental and demographic characteristics. I use a sample that includes only the control group sample from the two pre-reform years in these regressions. These results are presented in Online Appendix Table A.12. Using data for 13 years after high school graduation, I report in panel A estimates from regressions when only one of the high school outcomes is included in the regressions (columns 2) and estimates when all outcomes are included jointly in the regression (columns 3). When included one at a time, estimates of all high school outcomes are positive and very precisely measured. When all four are included jointly, all their estimates remain positive and statistically significant. These results are consistent with evidence reported in Lavy, Ebenstein, and Roth (2014) and Ebenstein, Lavy, and Roth (2016), who use random shocks to performance in matriculation exams to identify the reduced-form effects of these high school outcomes on earnings at adulthood, and find strong and significant positive effects.

In panel B, column 2, I report estimates from regressions of the enrolment and completed years of post-secondary schooling on earnings 13 years after high school graduation, first, when including only one of these outcomes at a time (column 2) and, second, where all four are included jointly (column 3). Here as well, each of these outcomes has a positive and significant association with earnings. In column 4, I report the estimates from a regression where all high school and post-secondary education outcomes are included jointly. Note that six of the eight outcomes still have positive and significant estimated coefficients. Three of the four high school outcomes are among them, even though the regression includes the post-secondary schooling variables. I view this as evidence that the high school outcomes that measure the quality of education affect earnings and their effect on post-secondary schooling. A second conclusion from Online Appendix Table A.9 is that high school and post-secondary schooling outcomes correlate with earnings.

Another way to check whether the program's effect on earnings stems from improvement in high school outcomes is to examine whether the estimated effect on post-secondary attainment and earnings shrinks or even disappears when the *Bagrut* outcomes are added as controls. Of course, such evidence could be only suggestive because the high school outcomes are endogenous and are probably correlated with the error term in the regressions of long-term outcomes. In Online Appendix Table A.13, I present estimates of the coefficient of the school choice treatment effect in a DID regression that also includes the high-school outcomes as explanatory variables—first, including one at a time and, second, including all four jointly. For ease of comparison, I present in column 1 the original treatment effect estimates from Tables 2 and 6. The effect of school choice on enrolment in college schooling is 0.049. The inclusion of each of the high school outcomes as an additional control shrinks the treatment effect by a third to a half toward zero. When all four high school outcomes are included, the effect is no longer statistically significant.

A similar pattern is seen in the second row of Online Appendix Table A.13 when the long-term outcome is completed years of college schooling. However, the most striking result is the sensitivity of the treatment effect on earnings to adding the high school outcomes as controls: the average matriculation score and the number of matriculation credits reduce the treatment effect from NIS 6,367 to about 1,800, a 70% decline.

To conclude, the analysis in this section suggests that there is evidence that the long-run effect of the program is driven both by a direct effect of the program on human capital and by an indirect effect of the program through earning a matriculation certificate and acceptance into post-secondary institutions. We caution once more that this analysis is only suggestive as the high-school outcomes and the post-secondary controls are endogenous.

7. Conclusions

The vast majority of published research on the impact of school interventions has examined their effects on short-run outcomes, primarily by looking at their impact on standardized test scores. While significant, a possible, more profound question is the impact of such interventions on life outcomes in the long run. This is a critical question because the ultimate goal of education is to improve lifetime well-being. Therefore, gaining new insights about which interventions are more effective in improving long-term outcomes will make a potent contribution to the design and implementation of new interventions, better resource allocation, and the education sector's efficiency.

Recent research has begun to look at this issue. Still, much work remains to be done, particularly concerning the long-term effects of interventions explicitly targeting improvement in the general quality of education and students' educational attainment. This study's empirical evidence contributes to a complete picture of the long-term returns to various educational interventions. This effort should enable teachers, institutions, and governments to make more informed decisions about which educational programs constitute the most beneficial use of limited school resources. The high school system in Israel and its high-stakes exit exams are similar to those in other countries. The school choice program studied in this paper shares many features with programs implemented in recent years in the United States and European and other OECD (Organisation for Economic Co-operation and Development) countries. As a result, the lessons learned from this research are transferable and applicable to the education system in other developed countries.

The school choice program studied here had positive longer-term outcomes in adulthood. The evidence suggests that allowing children to choose their secondary school freely at age 12 not only improved their high school outcomes sharply six years later, but it also impacted their path to post-secondary schooling and increased their earnings meaningfully about a decade and a half later. It is important to note that these gains were not at the expense of other students in the receiving schools. This latter group was already exposed to a similar proportion of incoming students due to the busing program that preceded the school choice program. If such a peer effect was operational, it should have been positive because opting out to schools outside the school district was voluntary in the choice program while being compulsory during the busing program. In earlier work, I have shown that improved outcomes during middle school and high school were facilitated by a better student–school match, more competition among schools, and higher schooling quality. These results are significant because the school choice experiment targeted a disadvantaged population in some of the more deprived parts of Tel-Aviv. Since evidence about the TASCP has become available, other school choice programs have been introduced, for example, in Jerusalem in 2006 and more recently in many cities in Israel.[36]

As a final remark, it is important to note that the TASCP is very similar to school-choice programs implemented in Europe, for example, in the Netherlands, Belgium, and Sweden, and in the United States, where free school choice programs have replaced, under court order, zoning and cross-district busing. But these programs, in Israel and elsewhere, are very different in fundamentals from charter schools and school voucher programs in the United States that are evaluated in the literature. For example, US voucher programs are targeted to students from small-income families, and charter schools are not "regular" public schools. They are exempt from many public education rules and regulations. The Tel-Aviv choice program highlights a compromise struck by the municipality between choice (among public schools) and regulation, which perhaps explains why the results presented in this paper are different from evidence in the United States, thus offering a path for other attempts at adjusting school choice programs.

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36. The Ministries of Education and Finance in Israel recently introduced a large school choice program in the largest cities in the country. As this recent expansion is reaching a nationwide scale, issues of general equilibrium effects become important, in particular with regard to whether the higher education system in Israel can accommodate the expected increased demand for post-secondary schooling. In this regard, it is important to note that the creation of academic colleges that started in the mid-1990s gained momentum and in the following two decades more such colleges were opened all over the country. This large supply expansion and the existing excess capacity in most of these colleges will be able to accommodate the increase in demand for higher education due to a country-wide school choice program, without repercussions for the existing demand.

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Supplementary Data

Supplementary data are available at *JEEA* online.